

# Kilonovae and *R*-process Nucleosynthesis from Neutron Star Mergers

DR. ATUL KEDIA

FEBRUARY 26, 2026

# Outline

## 1. Introduction

- Question: what is the astrophysical origin of *heavy* elements?
- Answer: Neutron star mergers?

## 2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

## 3. Modelling Astrophysics: How much *r*-process is produced? (mass)

## 4. Conclusions

# Outline

## 1. Introduction

- **Question: what is the astrophysical origin of *heavy* elements?**
- Answer: Neutron star mergers?

## 2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

## 3. Modelling Astrophysics: How much *r*-process is produced? (mass)

## 4. Conclusions

# The origin of elements

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                                      |                                 |                                         |                                   |                                      |                                    |                                   |                                    |                                  |                                    |                                     |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--------------------------------------|---------------------------------|-----------------------------------------|-----------------------------------|--------------------------------------|------------------------------------|-----------------------------------|------------------------------------|----------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|-----------------------------------|----------------------------------|------------------------------------|-----------------------------------|----------------------------------|--|
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13<br>IIIA                           |                                 | 14<br>IVA                               |                                   | 15<br>VA                             |                                    | 16<br>VIA                         |                                    | 17<br>VIIA                       |                                    | 18<br>VIIIA                         |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 5<br>B                               | 6<br>C                          | 7<br>N                                  | 8<br>O                            | 9<br>F                               | 10<br>Ne                           |                                   |                                    |                                  |                                    |                                     |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Boron<br>10.81<br>2.3                | Carbon<br>12.011<br>2.4         | Nitrogen<br>14.007<br>2.5               | Oxygen<br>15.999<br>2.6           | Fluorine<br>18.998<br>2.7            | Neon<br>20.180<br>2.8              |                                   |                                    |                                  |                                    |                                     |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13<br>Al                             | 14<br>Si                        | 15<br>P                                 | 16<br>S                           | 17<br>Cl                             | 18<br>Ar                           |                                   |                                    |                                  |                                    |                                     |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Aluminium<br>26.982<br>2.8           | Silicon<br>28.085<br>2.8        | Phosphorus<br>30.974<br>2.8             | Sulfur<br>32.06<br>2.8            | Chlorine<br>35.45<br>2.8             | Argon<br>39.948<br>2.8             |                                   |                                    |                                  |                                    |                                     |                                    |                                    |                                    |                                   |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 19<br>K                              | 20<br>Ca                        | 21<br>Sc                                | 22<br>Ti                          | 23<br>V                              | 24<br>Cr                           | 25<br>Mn                          | 26<br>Fe                           | 27<br>Co                         | 28<br>Ni                           | 29<br>Cu                            | 30<br>Zn                           | 31<br>Ga                           | 32<br>Ge                           | 33<br>As                          | 34<br>Se                         | 35<br>Br                           | 36<br>Kr                          |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Potassium<br>39.0983<br>2.8-4        | Calcium<br>40.078<br>2.8-2      | Scandium<br>44.955908<br>2.8-2          | Titanium<br>47.867<br>2.8-2       | Vanadium<br>50.9415<br>2.8-2         | Chromium<br>51.9961<br>2.8-3       | Manganese<br>54.938044<br>2.8-2   | Iron<br>55.845<br>2.8-2            | Cobalt<br>58.933<br>2.8-2        | Nickel<br>58.693<br>2.8-2          | Copper<br>63.546<br>2.8-2           | Zinc<br>65.38<br>2.8-2             | Gallium<br>69.723<br>2.8-3         | Germanium<br>72.630<br>2.8-4       | Arsenic<br>74.922<br>2.8-3        | Selenium<br>78.971<br>2.8-4      | Bromine<br>79.904<br>2.8-3         | Krypton<br>83.798<br>2.8-3        |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 37<br>Rb                             | 38<br>Sr                        | 39<br>Y                                 | 40<br>Zr                          | 41<br>Nb                             | 42<br>Mo                           | 43<br>Tc                          | 44<br>Ru                           | 45<br>Rh                         | 46<br>Pd                           | 47<br>Ag                            | 48<br>Cd                           | 49<br>In                           | 50<br>Sn                           | 51<br>Sb                          | 52<br>Te                         | 53<br>I                            | 54<br>Xe                          |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rubidium<br>85.4678<br>2.8-36-1      | Strontium<br>87.62<br>2.8-2     | Yttrium<br>88.90584<br>2.8-2            | Zirconium<br>91.224<br>2.8-3-2    | Niobium<br>92.90637<br>2.8-3-2       | Molybdenum<br>95.95<br>2.8-3-2     | Technetium<br>98<br>2.8-3-2       | Ruthenium<br>101.07<br>2.8-3-1     | Rhodium<br>102.91<br>2.8-3-1     | Palladium<br>106.42<br>2.8-3-2     | Silver<br>107.87<br>2.8-3-1         | Cadmium<br>112.41<br>2.8-3-2       | Indium<br>114.82<br>2.8-3-3        | Tin<br>118.71<br>2.8-3-4           | Antimony<br>121.76<br>2.8-3-3     | Tellurium<br>127.60<br>2.8-3-4   | Iodine<br>126.90<br>2.8-3-4        | Xenon<br>131.29<br>2.8-3-4        |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 55<br>Cs                             | 56<br>Ba                        | 57-71<br>Lanthanides                    |                                   | 72<br>Hf                             | 73<br>Ta                           | 74<br>W                           | 75<br>Re                           | 76<br>Os                         | 77<br>Ir                           | 78<br>Pt                            | 79<br>Au                           | 80<br>Hg                           | 81<br>Tl                           | 82<br>Pb                          | 83<br>Bi                         | 84<br>Po                           | 85<br>At                          | 86<br>Rn                         |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Cesium<br>132.90545196<br>2.8-36-8-1 | Barium<br>137.327<br>2.8-36-1   |                                         |                                   | Hafnium<br>178.49<br>2.8-3-3-2       | Tantalum<br>180.94788<br>2.8-3-3-2 | Tungsten<br>183.84<br>2.8-3-3-2   | Rhenium<br>186.21<br>2.8-3-3-2     | Osmium<br>190.23<br>2.8-3-3-2    | Iridium<br>192.22<br>2.8-3-3-2     | Platinum<br>195.08<br>2.8-3-3-1     | Gold<br>196.97<br>2.8-3-3-1        | Mercury<br>200.59<br>2.8-3-3-2     | Thallium<br>204.38<br>2.8-3-3-3    | Lead<br>207.2<br>2.8-3-3-4        | Bismuth<br>208.98<br>2.8-3-3-4   | Polonium<br>209<br>2.8-3-3-4       | Astatine<br>210<br>2.8-3-3-4      | Radon<br>222<br>2.8-3-3-4        |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 87<br>Fr                             | 88<br>Ra                        | 89-103<br>Actinides                     |                                   | 104<br>Rf                            | 105<br>Db                          | 106<br>Sg                         | 107<br>Bh                          | 108<br>Hs                        | 109<br>Mt                          | 110<br>Ds                           | 111<br>Rg                          | 112<br>Cn                          | 113<br>Nh                          | 114<br>Fl                         | 115<br>Mc                        | 116<br>Lv                          | 117<br>Ts                         | 118<br>Og                        |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Francium<br>223<br>2.8-3-37-8-1      | Radium<br>226<br>2.8-3-37-8-1   |                                         |                                   | Rutherfordium<br>261<br>2.8-3-32-3-2 | Dubnium<br>262<br>2.8-3-32-3-2     | Seaborgium<br>266<br>2.8-3-32-3-2 | Bohrium<br>264<br>2.8-3-32-3-2     | Hassium<br>277<br>2.8-3-32-3-2   | Meitnerium<br>268<br>2.8-3-32-3-2  | Darmstadtium<br>271<br>2.8-3-32-3-1 | Roentgenium<br>272<br>2.8-3-32-3-2 | Copernicium<br>285<br>2.8-3-32-3-2 | Nihonium<br>284<br>2.8-3-32-3-2    | Flerovium<br>289<br>2.8-3-32-3-4  | Moscovium<br>288<br>2.8-3-32-3-4 | Livermorium<br>293<br>2.8-3-32-3-4 | Tennessine<br>294<br>2.8-3-32-3-8 | Oganesson<br>294<br>2.8-3-32-3-8 |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 57<br>La                             | 58<br>Ce                        | 59<br>Pr                                | 60<br>Nd                          | 61<br>Pm                             | 62<br>Sm                           | 63<br>Eu                          | 64<br>Gd                           | 65<br>Tb                         | 66<br>Dy                           | 67<br>Ho                            | 68<br>Er                           | 69<br>Tm                           | 70<br>Yb                           | 71<br>Lu                          |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Lanthanum<br>138.905<br>2.8-36-3-2   | Cerium<br>140.12<br>2.8-36-3-2  | Praseodymium<br>140.90766<br>2.8-36-3-2 | Neodymium<br>144.24<br>2.8-36-3-2 | Promethium<br>144.9127<br>2.8-36-3-2 | Samarium<br>150.36<br>2.8-36-3-2   | Europium<br>151.964<br>2.8-36-3-2 | Gadolinium<br>157.25<br>2.8-36-3-2 | Terbium<br>158.925<br>2.8-36-3-2 | Dysprosium<br>162.50<br>2.8-36-3-2 | Holmium<br>164.93032<br>2.8-36-3-2  | Erbium<br>167.259<br>2.8-36-3-2    | Thulium<br>168.934<br>2.8-36-3-2   | Ytterbium<br>173.054<br>2.8-36-3-2 | Lutetium<br>174.967<br>2.8-36-3-2 |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 89<br>Ac                             | 90<br>Th                        | 91<br>Pa                                | 92<br>U                           | 93<br>Np                             | 94<br>Pu                           | 95<br>Am                          | 96<br>Cm                           | 97<br>Bk                         | 98<br>Cf                           | 99<br>Es                            | 100<br>Fm                          | 101<br>Md                          | 102<br>No                          | 103<br>Lr                         |                                  |                                    |                                   |                                  |  |
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Actinium<br>227<br>2.8-36-3-2        | Thorium<br>232.04<br>2.8-36-3-2 | Protactinium<br>231<br>2.8-36-3-2       | Uranium<br>238.03<br>2.8-36-3-2   | Neptunium<br>237<br>2.8-36-3-2       | Plutonium<br>244<br>2.8-36-3-2     | Americium<br>243<br>2.8-36-3-2    | Curium<br>247<br>2.8-36-3-2        | Berkelium<br>247<br>2.8-36-3-2   | Californium<br>251<br>2.8-36-3-2   | Einsteinium<br>252<br>2.8-36-3-2    | Fermium<br>257<br>2.8-36-3-2       | Mendelevium<br>258<br>2.8-36-3-2   | Nobelium<br>259<br>2.8-36-3-2      | Lawrencium<br>262<br>2.8-36-3-3   |                                  |                                    |                                   |                                  |  |

State of matter (color of name)  
GAS LIQUID SOLID UNKNOWN

Subcategory in the metal-metalloid-nonmetal trend (color of background)  
Alkali metals Lanthanides Metalloids  
Alkaline earth metals Actinides Reactive nonmetals  
Transition metals Post-transition metals Noble gases

Unknown chemical properties



# The origin of elements

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                |                                                              |                                                             |                                                            |                                                           |                                                             |                                                               |                                                              |                                                              |                                                             |                                                            |                                                              |                                                              |                                                             |                                                            |  |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|--|
| <div><div>H<sup>1</sup></div><div>Hydrogen</div></div>   | <div><div></div><div>The big bang</div></div> <div><div></div><div>Dying low-mass stars</div></div> <div><div></div><div>White dwarf supernovae</div></div> <div><div></div><div>Radioactive decay</div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                |                                                              |                                                             |                                                            |                                                           |                                                             |                                                               |                                                              |                                                              |                                                             |                                                            |                                                              |                                                              |                                                             | <div><div>He<sup>2</sup></div><div>Helium</div></div>      |  |
| <div><div>Li<sup>3</sup></div><div>Lithium</div></div>   | <div><div>Be<sup>4</sup></div><div>Beryllium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                |                                                              |                                                             |                                                            |                                                           |                                                             |                                                               |                                                              |                                                              |                                                             |                                                            |                                                              |                                                              |                                                             |                                                            |  |
| <div><div>Na<sup>11</sup></div><div>Sodium</div></div>   | <div><div>Mg<sup>12</sup></div><div>Magnesium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div><div></div><div>Cosmic ray collisions</div></div> <div><div></div><div>Dying high-mass stars</div></div> <div><div></div><div>Merging neutron stars</div></div> <div><div></div><div>Human-made</div></div> |                                                                |                                                              |                                                             |                                                            |                                                           |                                                             |                                                               |                                                              |                                                              |                                                             |                                                            |                                                              |                                                              |                                                             |                                                            |  |
| <div><div>K<sup>19</sup></div><div>Potassium</div></div> | <div><div>Ca<sup>20</sup></div><div>Calcium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <div><div>Sc<sup>21</sup></div><div>Scandium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <div><div>Ti<sup>22</sup></div><div>Titanium</div></div>       | <div><div>V<sup>23</sup></div><div>Vanadium</div></div>      | <div><div>Cr<sup>24</sup></div><div>Chromium</div></div>    | <div><div>Mn<sup>25</sup></div><div>Manganese</div></div>  | <div><div>Fe<sup>26</sup></div><div>Iron</div></div>      | <div><div>Co<sup>27</sup></div><div>Cobalt</div></div>      | <div><div>Ni<sup>28</sup></div><div>Nickel</div></div>        | <div><div>Cu<sup>29</sup></div><div>Copper</div></div>       | <div><div>Zn<sup>30</sup></div><div>Zinc</div></div>         | <div><div>Ga<sup>31</sup></div><div>Gallium</div></div>     | <div><div>Ge<sup>32</sup></div><div>Germanium</div></div>  | <div><div>As<sup>33</sup></div><div>Arsenic</div></div>      | <div><div>Se<sup>34</sup></div><div>Selenium</div></div>     | <div><div>Br<sup>35</sup></div><div>Bromine</div></div>     | <div><div>Kr<sup>36</sup></div><div>Krypton</div></div>    |  |
| <div><div>Rb<sup>37</sup></div><div>Rubidium</div></div> | <div><div>Sr<sup>38</sup></div><div>Strontium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div><div>Y<sup>39</sup></div><div>Yttrium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <div><div>Zr<sup>40</sup></div><div>Zirconium</div></div>      | <div><div>Nb<sup>41</sup></div><div>Niobium</div></div>      | <div><div>Mo<sup>42</sup></div><div>Molybdenum</div></div>  | <div><div>Tc<sup>43</sup></div><div>Technetium</div></div> | <div><div>Ru<sup>44</sup></div><div>Ruthenium</div></div> | <div><div>Rh<sup>45</sup></div><div>Rhodium</div></div>     | <div><div>Pd<sup>46</sup></div><div>Palladium</div></div>     | <div><div>Ag<sup>47</sup></div><div>Silver</div></div>       | <div><div>Cd<sup>48</sup></div><div>Cadmium</div></div>      | <div><div>In<sup>49</sup></div><div>Indium</div></div>      | <div><div>Sn<sup>50</sup></div><div>Tin</div></div>        | <div><div>Sb<sup>51</sup></div><div>Antimony</div></div>     | <div><div>Te<sup>52</sup></div><div>Tellurium</div></div>    | <div><div>I<sup>53</sup></div><div>Iodine</div></div>       | <div><div>Xe<sup>54</sup></div><div>Xenon</div></div>      |  |
| <div><div>Cs<sup>55</sup></div><div>Cesium</div></div>   | <div><div>Ba<sup>56</sup></div><div>Barium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div><div>Hf<sup>72</sup></div><div>Hafnium</div></div>        | <div><div>Ta<sup>73</sup></div><div>Tantalum</div></div>     | <div><div>W<sup>74</sup></div><div>Tungsten</div></div>     | <div><div>Re<sup>75</sup></div><div>Rhenium</div></div>    | <div><div>Os<sup>76</sup></div><div>Osmium</div></div>    | <div><div>Ir<sup>77</sup></div><div>Iridium</div></div>     | <div><div>Pt<sup>78</sup></div><div>Platinum</div></div>      | <div><div>Au<sup>79</sup></div><div>Gold</div></div>         | <div><div>Hg<sup>80</sup></div><div>Mercury</div></div>      | <div><div>Tl<sup>81</sup></div><div>Thallium</div></div>    | <div><div>Pb<sup>82</sup></div><div>Lead</div></div>       | <div><div>Bi<sup>83</sup></div><div>Bismuth</div></div>      | <div><div>Po<sup>84</sup></div><div>Polonium</div></div>     | <div><div>At<sup>85</sup></div><div>Astatine</div></div>    | <div><div>Rn<sup>86</sup></div><div>Radon</div></div>      |  |
| <div><div>Fr<sup>87</sup></div><div>Francium</div></div> | <div><div>Ra<sup>88</sup></div><div>Radium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div><div>Rf<sup>104</sup></div><div>Rutherfordium</div></div> | <div><div>Db<sup>105</sup></div><div>Dubnium</div></div>     | <div><div>Sg<sup>106</sup></div><div>Seaborgium</div></div> | <div><div>Bh<sup>107</sup></div><div>Bohrium</div></div>   | <div><div>Hs<sup>108</sup></div><div>Hassium</div></div>  | <div><div>Mt<sup>109</sup></div><div>Meitnerium</div></div> | <div><div>Ds<sup>110</sup></div><div>Darmstadtium</div></div> | <div><div>Rg<sup>111</sup></div><div>Roentgenium</div></div> | <div><div>Cn<sup>112</sup></div><div>Copernicium</div></div> | <div><div>Nh<sup>113</sup></div><div>Nihonium</div></div>   | <div><div>Fl<sup>114</sup></div><div>Flerovium</div></div> | <div><div>Mc<sup>115</sup></div><div>Moscovium</div></div>   | <div><div>Lv<sup>116</sup></div><div>Livermorium</div></div> | <div><div>Ts<sup>117</sup></div><div>Tennessine</div></div> | <div><div>Og<sup>118</sup></div><div>Oganesson</div></div> |  |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div><div>La<sup>57</sup></div><div>Lanthanum</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <div><div>Ce<sup>58</sup></div><div>Cerium</div></div>         | <div><div>Pr<sup>59</sup></div><div>Praseodymium</div></div> | <div><div>Nd<sup>60</sup></div><div>Neodymium</div></div>   | <div><div>Pm<sup>61</sup></div><div>Promethium</div></div> | <div><div>Sm<sup>62</sup></div><div>Samarium</div></div>  | <div><div>Eu<sup>63</sup></div><div>Europium</div></div>    | <div><div>Gd<sup>64</sup></div><div>Gadolinium</div></div>    | <div><div>Tb<sup>65</sup></div><div>Terbium</div></div>      | <div><div>Dy<sup>66</sup></div><div>Dysprosium</div></div>   | <div><div>Ho<sup>67</sup></div><div>Holmium</div></div>     | <div><div>Er<sup>68</sup></div><div>Erbium</div></div>     | <div><div>Tm<sup>69</sup></div><div>Thulium</div></div>      | <div><div>Yb<sup>70</sup></div><div>Ytterbium</div></div>    | <div><div>Lu<sup>71</sup></div><div>Lutetium</div></div>    |                                                            |  |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div><div>Ac<sup>89</sup></div><div>Actinium</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <div><div>Th<sup>90</sup></div><div>Thorium</div></div>        | <div><div>Pa<sup>91</sup></div><div>Protactinium</div></div> | <div><div>U<sup>92</sup></div><div>Uranium</div></div>      | <div><div>Np<sup>93</sup></div><div>Neptunium</div></div>  | <div><div>Pu<sup>94</sup></div><div>Plutonium</div></div> | <div><div>Am<sup>95</sup></div><div>Americium</div></div>   | <div><div>Cm<sup>96</sup></div><div>Curium</div></div>        | <div><div>Bk<sup>97</sup></div><div>Berkelium</div></div>    | <div><div>Cf<sup>98</sup></div><div>Californium</div></div>  | <div><div>Es<sup>99</sup></div><div>Einsteinium</div></div> | <div><div>Fm<sup>100</sup></div><div>Fermium</div></div>   | <div><div>Md<sup>101</sup></div><div>Mendelevium</div></div> | <div><div>No<sup>102</sup></div><div>Nobelium</div></div>    | <div><div>Lr<sup>103</sup></div><div>Lawrencium</div></div> |                                                            |  |

## NASA's Goddard Space Flight Center

# Density of objects in the Universe

Water



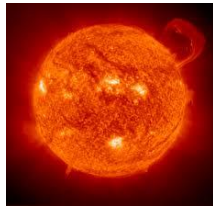
$$1 \text{ g cm}^{-3}$$

Earth



$$2 - 15 \text{ g cm}^{-3}$$

Sun



$$10 - 200 \text{ g cm}^{-3}$$

Neutron Star



$$? \text{ g cm}^{-3}$$

# Density of objects in the Universe

A.  $1,000 \text{ g cm}^{-3}$

B.  $100,000 \text{ g cm}^{-3}$

C.  $10,000,000 \text{ g cm}^{-3}$

D.  $1,000,000,000,000,000 \text{ g cm}^{-3} \sim 10^{15}$  ✓

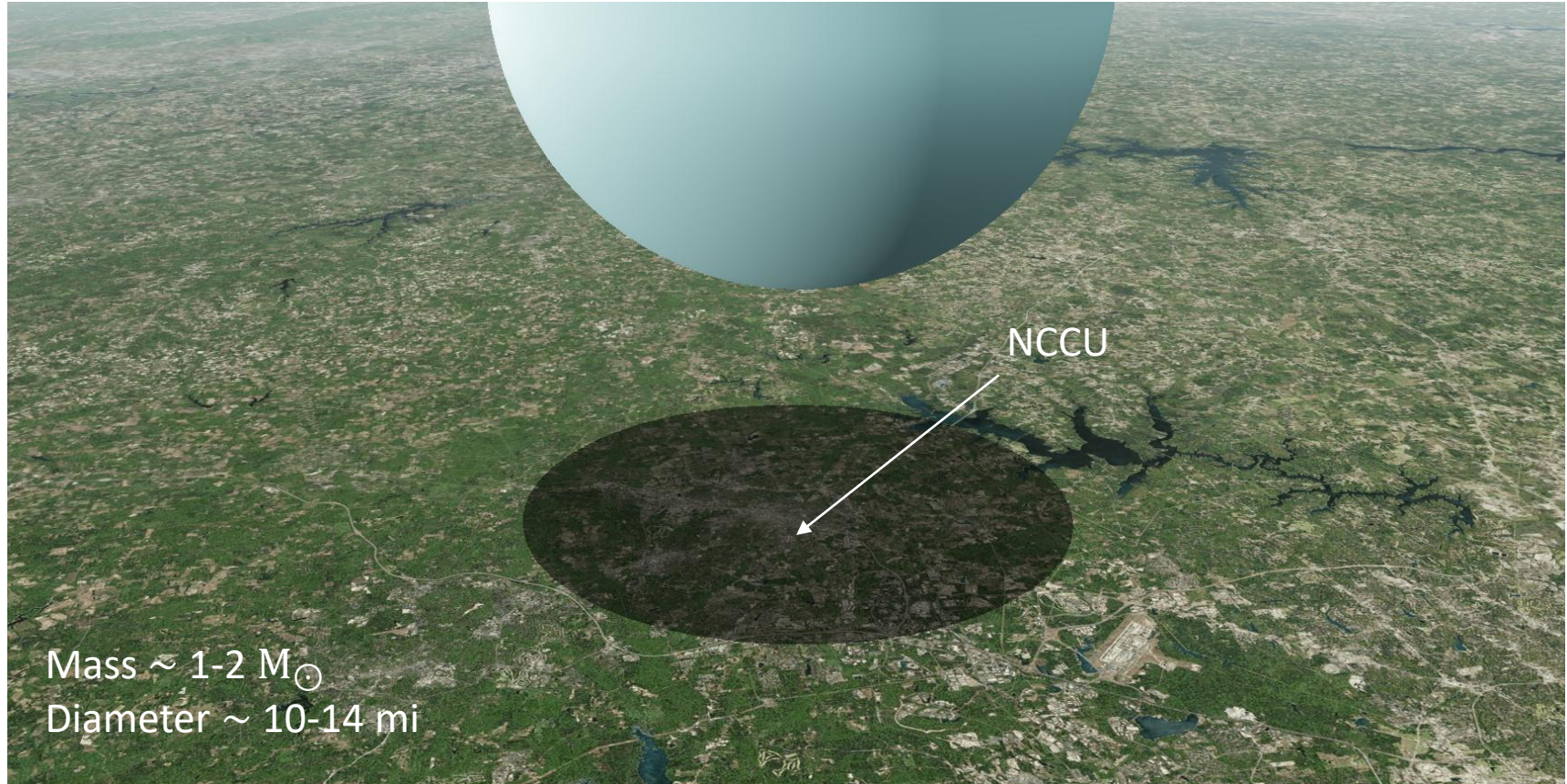
Neutron Star



$10^{15} \text{ g cm}^{-3}$



# Neutron star over Durham





# Neutron star merger



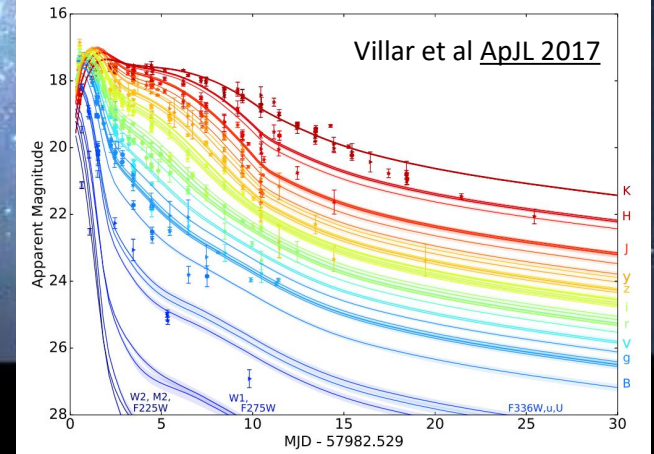
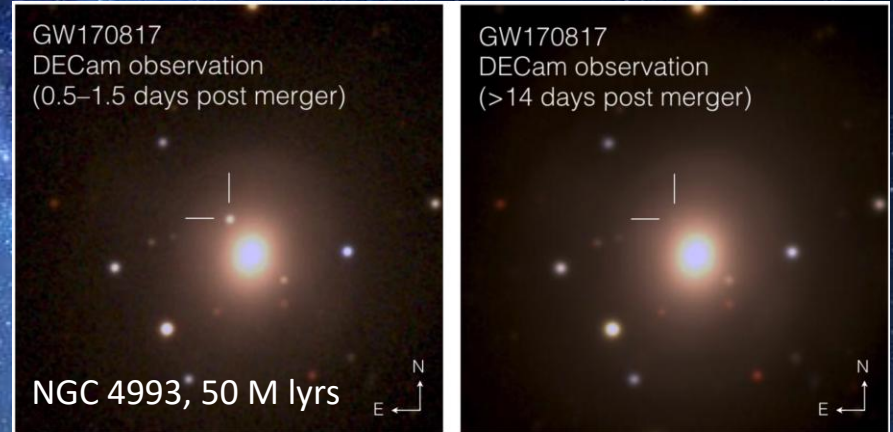
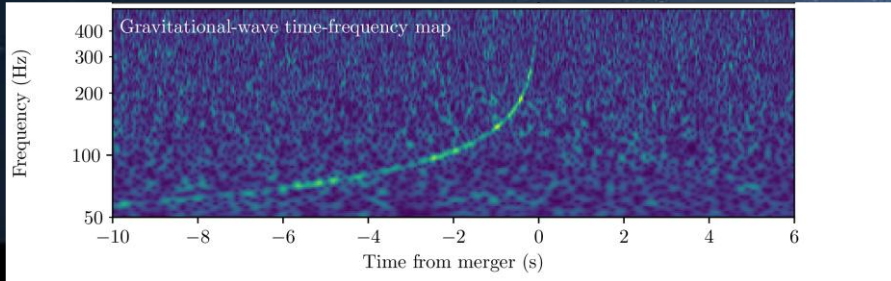
Artistic illustration from [NASA Goddard](#)

# Neutron star merger: GW170817/AT2017gfo



Two perpendicular 2.5 mile arms making  
a Michelson-Morley setup

LIGO-Livingston-Louisiana, LIGO-Hanford-Washington,  
Virgo-Italy, KARGA-Japan, upcoming LIGO-India (2030s)



# Modelling a neutron star merger

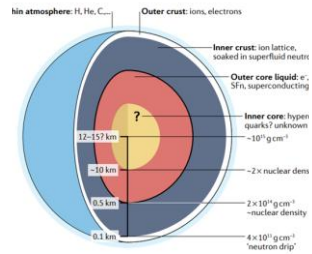
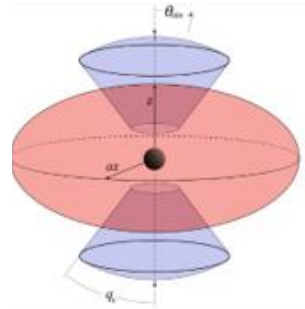


Fig. 1 | Schematic of the structure of a neutron star and its internal structure. The diagram illustrates the thin atmosphere, the outer and inner crust, and the outer and inner core, with the respective densities at different depths. Adapted with permission from the NICER Team.

## Neutron star structure

Equation of state

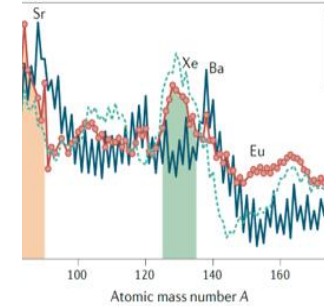
Bovard+17, Radice+18, Prakash+22,  
**Kedia+22, Kedia+24, Hensh+24**



## Hydrodynamics

General Relativistic

Breschi+21, Bulla+19,23, Heinzl+21,  
Kawaguchi+20,21, **Kedia+23, Fryer+24**



## Nuclear reactions

*r*-process nucleosynthesis

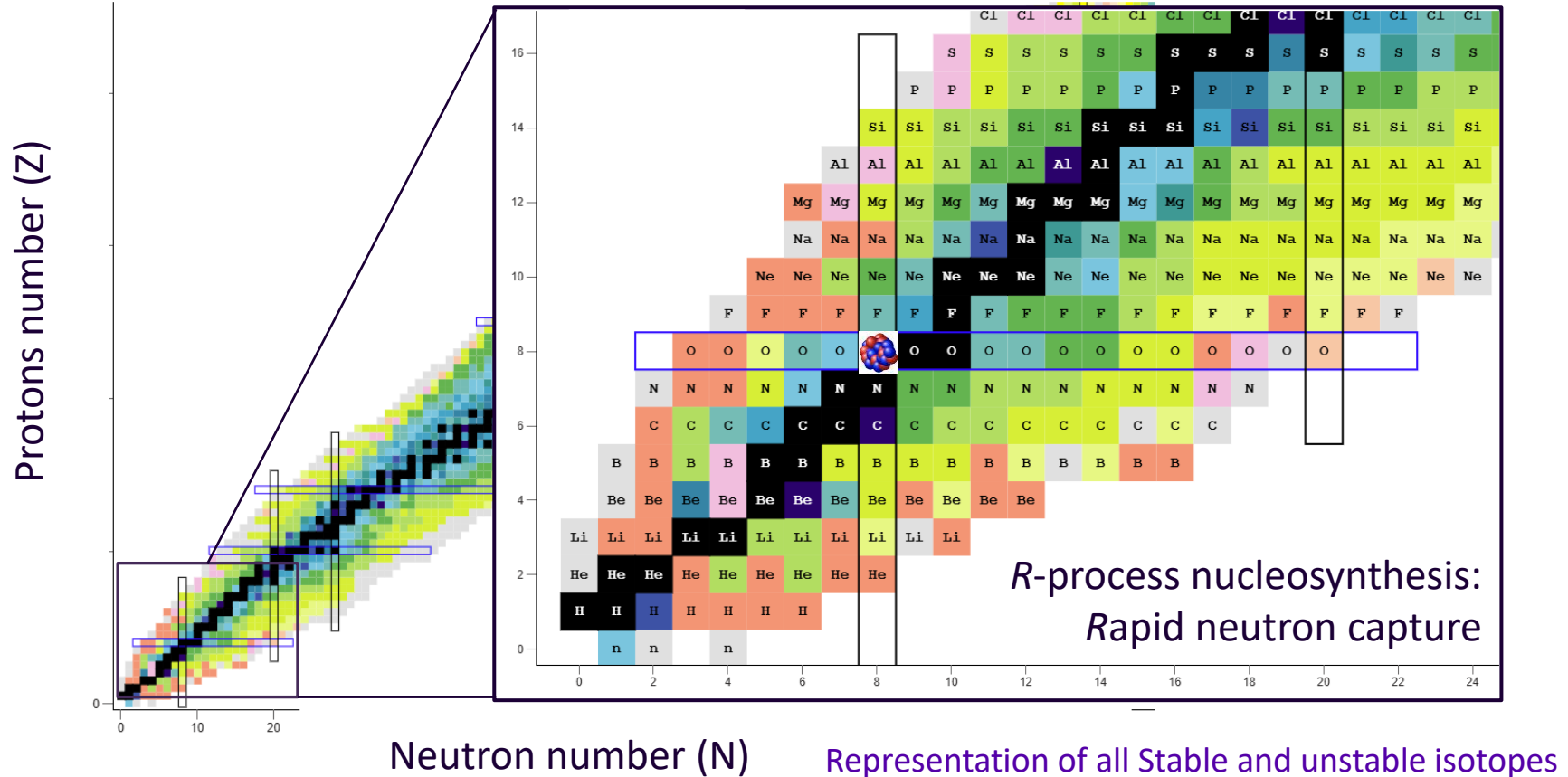
Mumpower+16, Vassh+18,+20,  
Zhu+18,Holmbeck+23, **Kedia+26**

# Outline

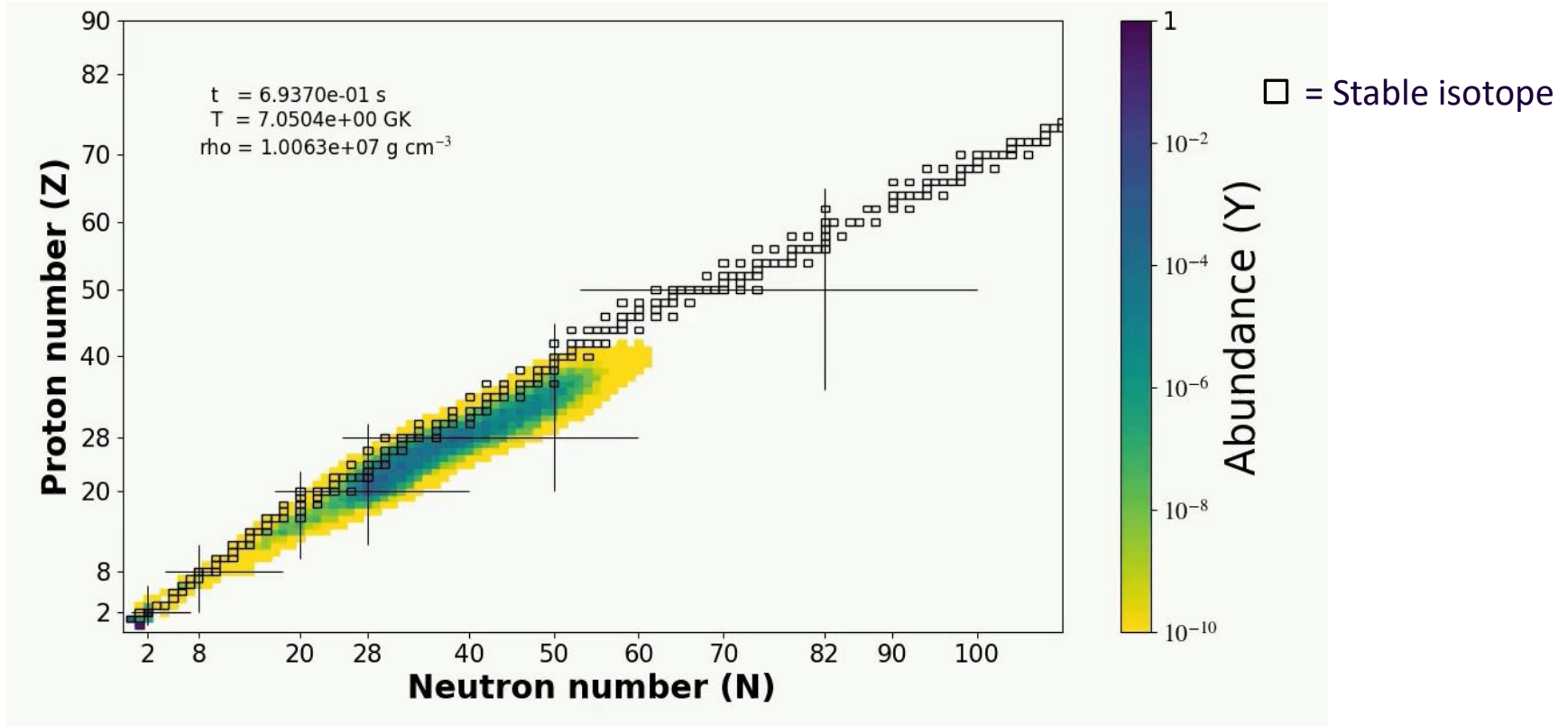
1. Introduction
  - Question: what is the astrophysical origin of *heavy* elements?
  - Answer: Neutron star mergers?
2. **Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)**
3. Modelling Astrophysics: How much *r*-process is produced? (mass)
4. Conclusions



# R-process on the Chart of Nuclides

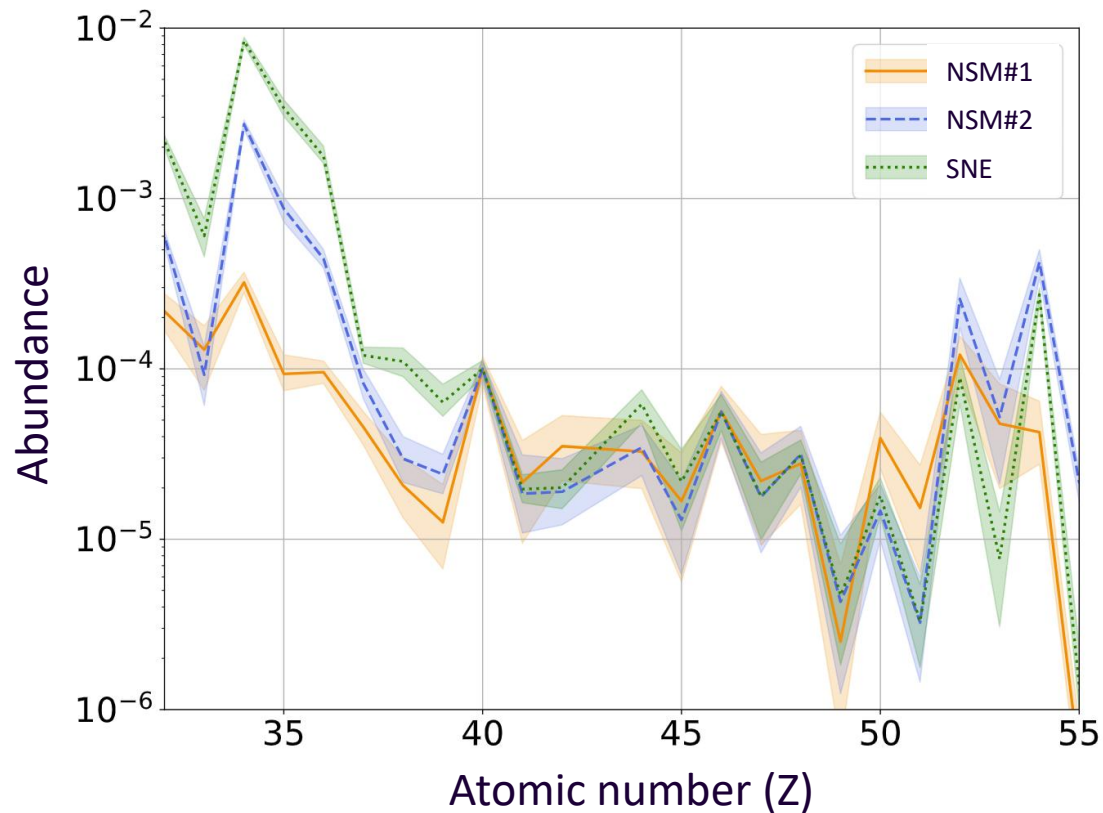


# R-process evolution



Movie by AK; Calculation done on **PRISM** (reaction network code)

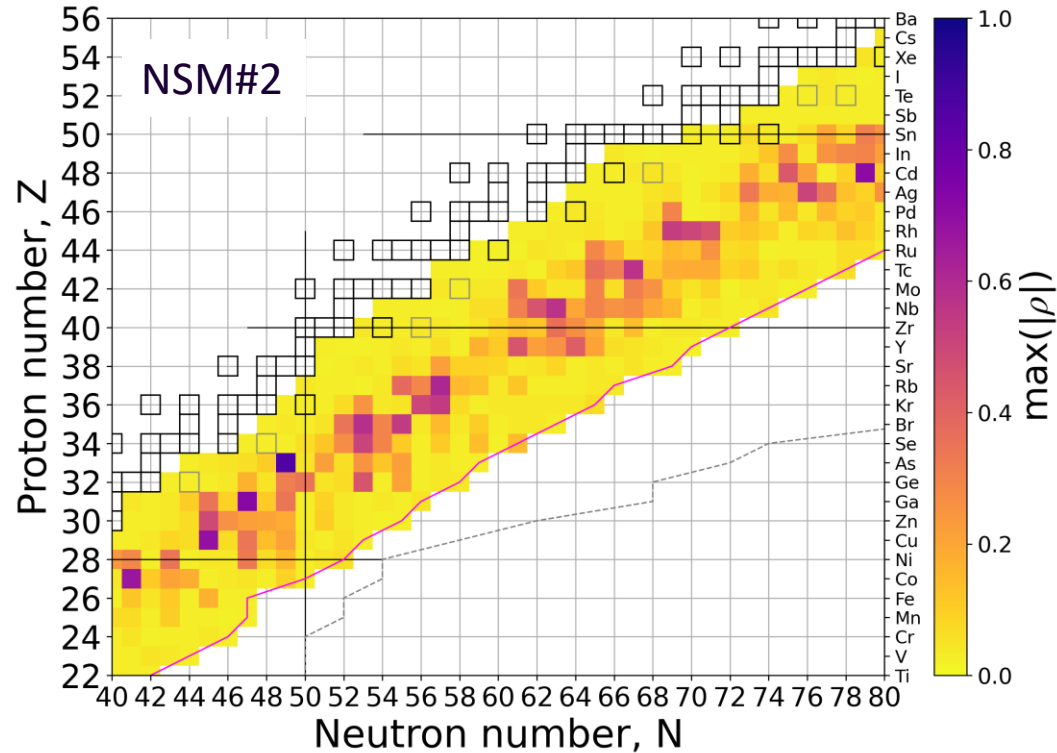
# Uncertainties in $r$ -process abundances



$2\sigma$  (95 percentile) uncertainty in abundances due to uncertainties in theoretical neutron capture reaction rates.

\*Shown is only the weak  $r$ -process abundance pattern.

# Most influential isotopes for $r$ process

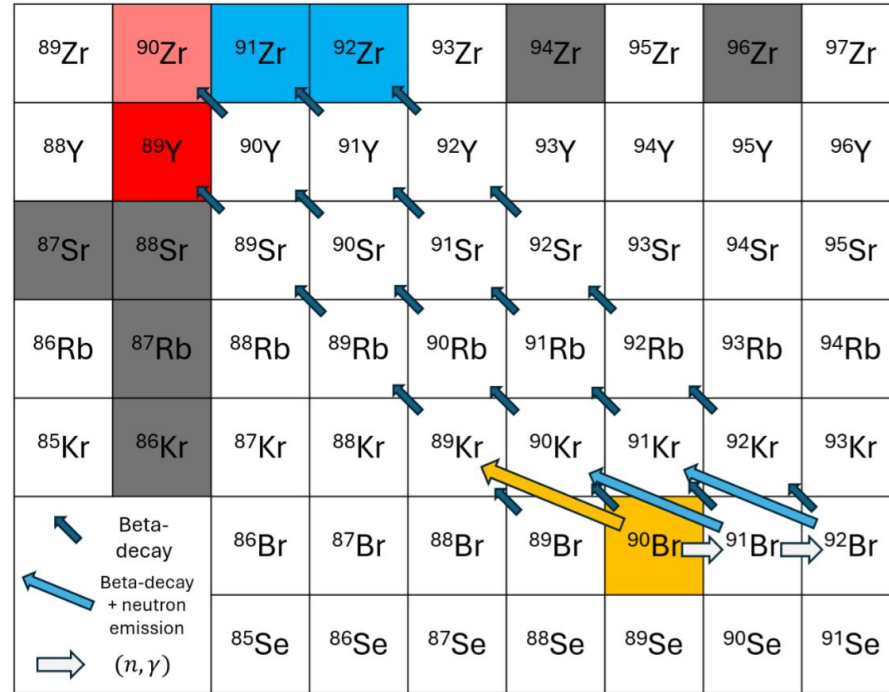


| Element | (n, $\gamma$ ) isotope, correlation coeff. (MF14) |                         |                         |   |
|---------|---------------------------------------------------|-------------------------|-------------------------|---|
| Se      | $^{78}\text{Ga}$ -0.65                            | $^{80}\text{Ga}$ -0.33  | -                       | - |
| Br      | $^{78}\text{Ga}$ 0.72                             | $^{80}\text{Ga}$ 0.34   | $^{81}\text{Ge}$ -0.32  | - |
| Kr      | $^{82}\text{As}$ 0.86                             | $^{82}\text{Ge}$ 0.35   | -                       | - |
| Rb      | $^{87}\text{Se}$ -0.51                            | $^{85}\text{Ge}$ -0.45  | -                       | - |
| Sr      | $^{88}\text{Br}$ -0.58                            | $^{87}\text{Br}$ 0.35   | -                       | - |
| Y       | $^{90}\text{Br}$ -0.54                            | $^{89}\text{Br}$ -0.30  | -                       | - |
| Zr      | $^{92}\text{Kr}$ -0.47                            | $^{94}\text{Rb}$ -0.40  | -                       | - |
| Nb      | $^{93}\text{Kr}$ -0.54                            | $^{92}\text{Kr}$ 0.50   | $^{92}\text{Rb}$ 0.38   | - |
| Mo      | $^{94}\text{Rb}$ 0.61                             | $^{100}\text{Y}$ -0.45  | -                       | - |
| Ru      | $^{104}\text{Nb}$ -0.57                           | -                       | -                       | - |
| Rh      | $^{103}\text{Nb}$ -0.45                           | $^{103}\text{Y}$ -0.42  | $^{103}\text{Zr}$ -0.36 | - |
| Pd      | $^{110}\text{Tc}$ -0.47                           | $^{104}\text{Nb}$ 0.34  | -                       | - |
| Ag      | $^{108}\text{Tc}$ 0.37                            | $^{109}\text{Tc}$ -0.32 | $^{109}\text{Mo}$ -0.32 | - |
| Cd      | $^{110}\text{Tc}$ 0.57                            | -                       | -                       | - |
| In      | $^{114}\text{Rh}$ 0.53                            | $^{115}\text{Rh}$ -0.50 | -                       | - |
| Sn      | $^{116}\text{Rh}$ 0.46                            | $^{124}\text{Ag}$ -0.34 | -                       | - |
| Sb      | $^{123}\text{Ag}$ -0.51                           | $^{123}\text{Cd}$ -0.44 | $^{120}\text{Ag}$ 0.33  | - |
| Te      | $^{130}\text{In}$ -0.62                           | $^{129}\text{Cd}$ 0.33  | -                       | - |
| I       | $^{127}\text{Cd}$ -0.71                           | -                       | -                       | - |
| Xe      | $^{130}\text{In}$ 0.62                            | $^{129}\text{Cd}$ -0.38 | $^{129}\text{Sn}$ -0.32 | - |



# Yttrium – $^{90}\text{Br}$

$^{90}\text{Br}$  has a 25% probability of beta-delayed neutron emission and indirectly impact **Yttrium** formation



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

# Reduction in uncertainties

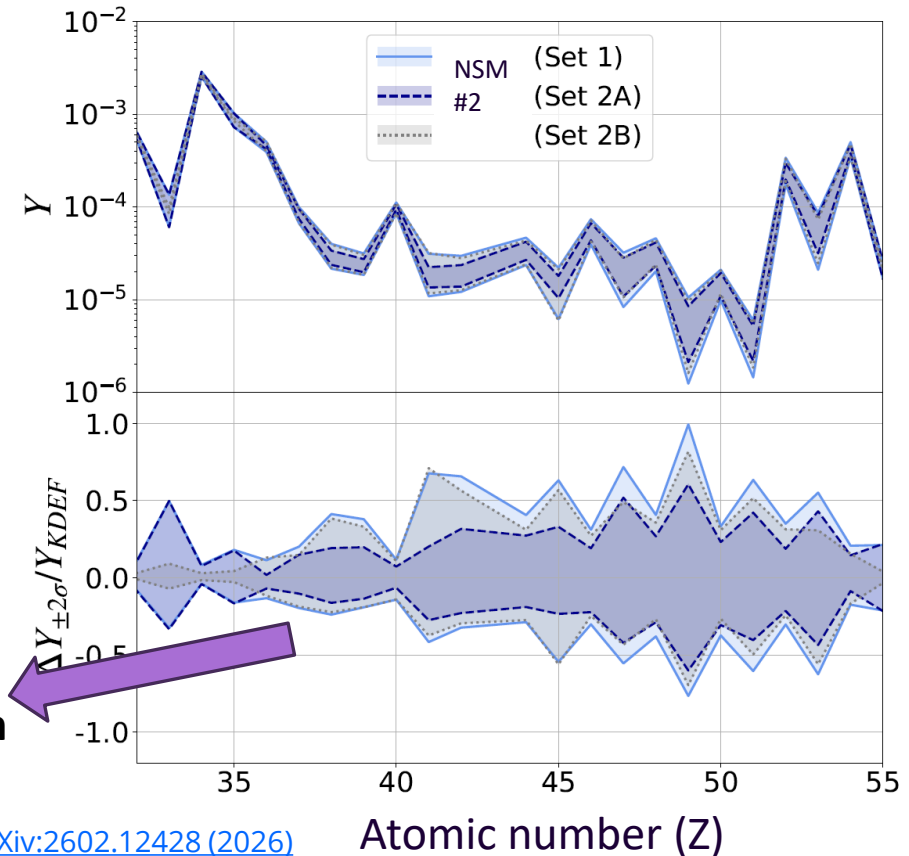
$2\sigma$  (95 percentile) uncertainty bands  
**Hypothetical reduction in uncertainties**

Set 1 : No reduction

Set 2A: Reduce uncertainties on rates for  
only **50** isotopes

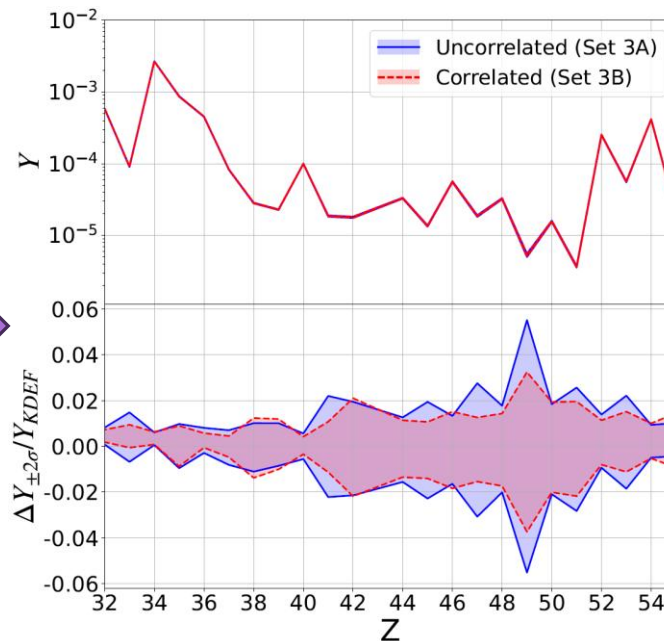
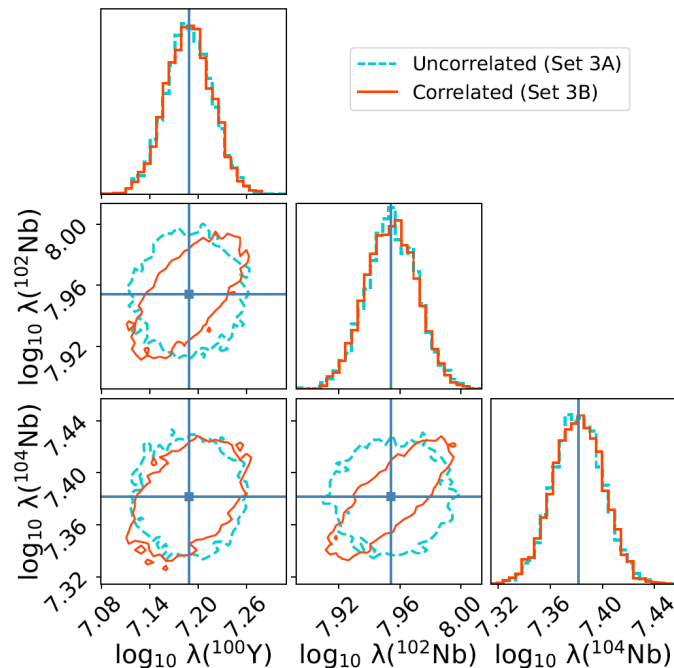
Set 2B: Reduce uncertainties on rates for  
all **1000** but not for the 50  
isotopes.

**The 50 isotopes lead to more reduction in  
uncertainty than the remaining 1000 do!!**



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

# Theoretical constraints of Reaction rates



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

# Outline

## 1. Introduction

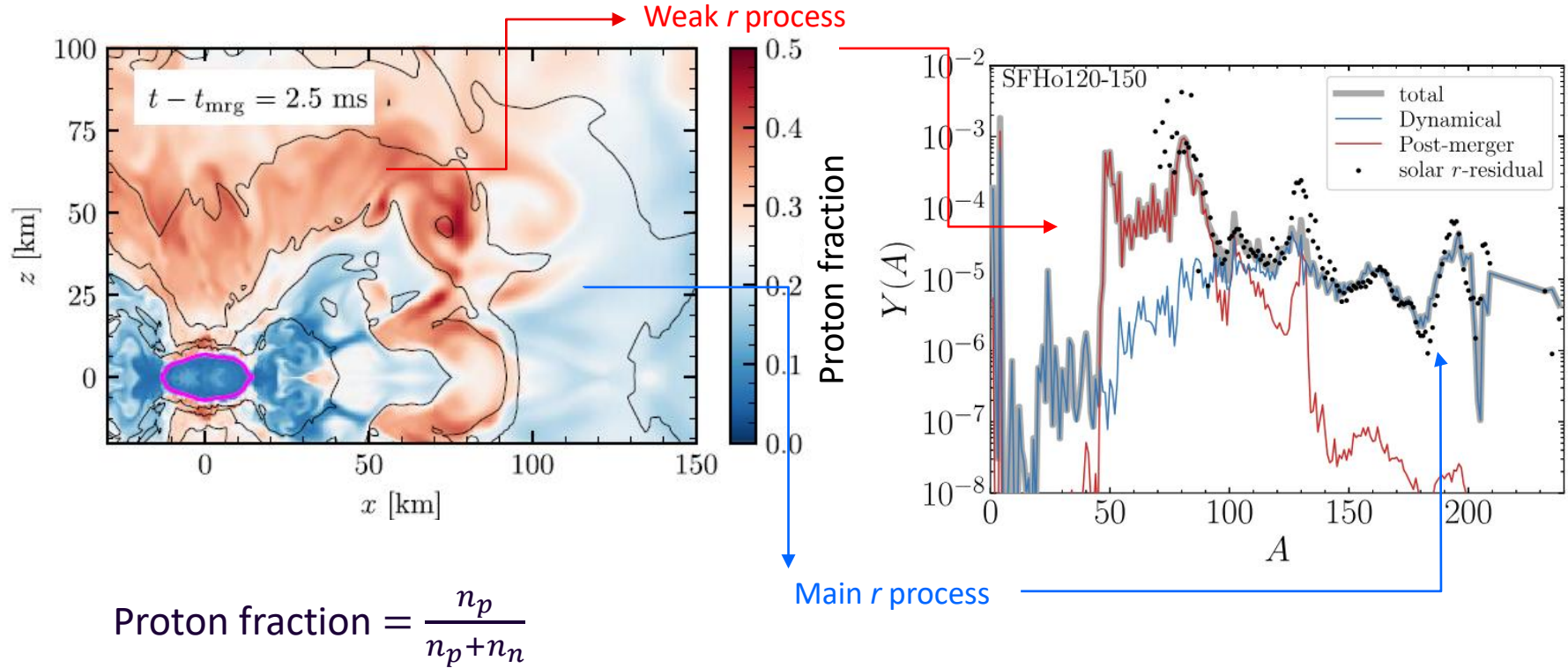
- Question: what is the astrophysical origin of *heavy* elements?
- Answer: Neutron star mergers?

## 2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

## 3. **Modelling Astrophysics: How much *r*-process is produced? (mass)**

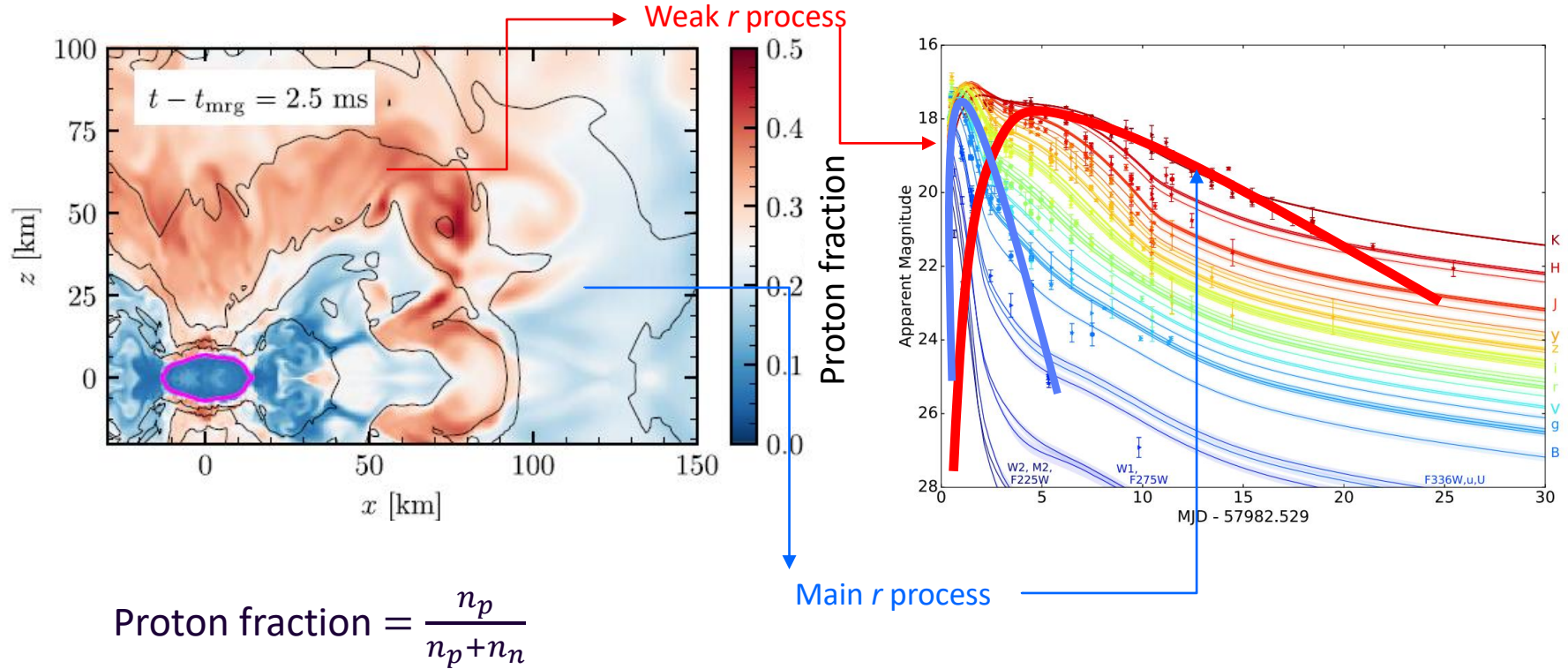
## 4. Conclusions

# Abundance dependence on ejecta

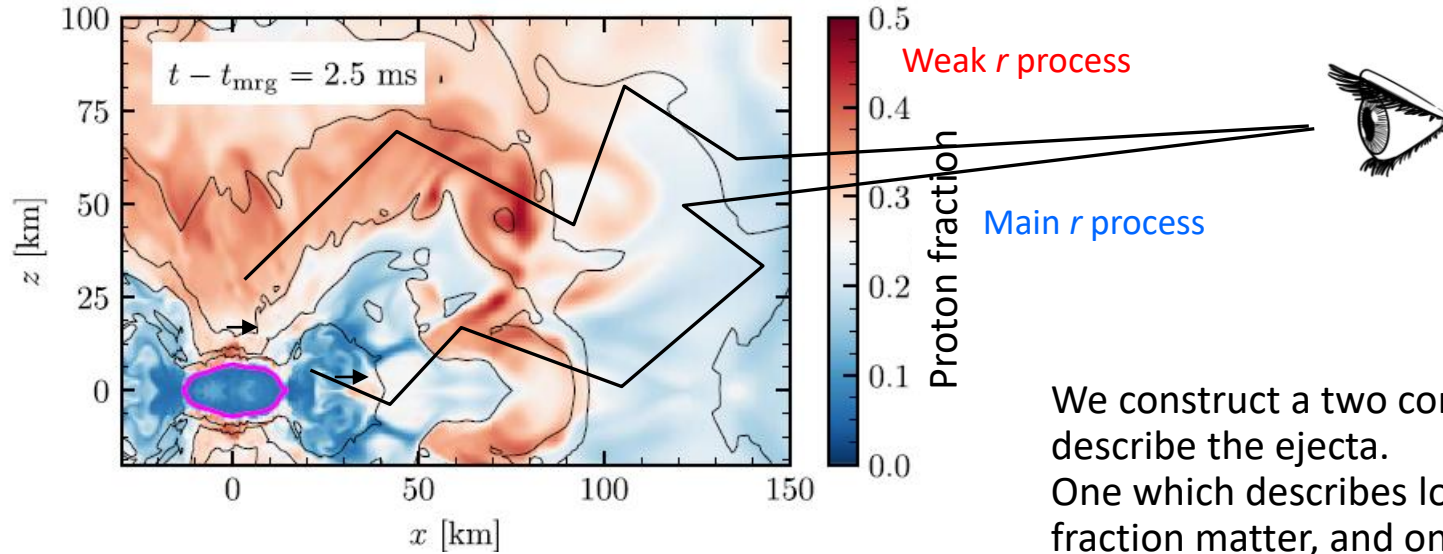




# Kilonova contribution of the ejecta



# Kilonova contribution of the ejecta

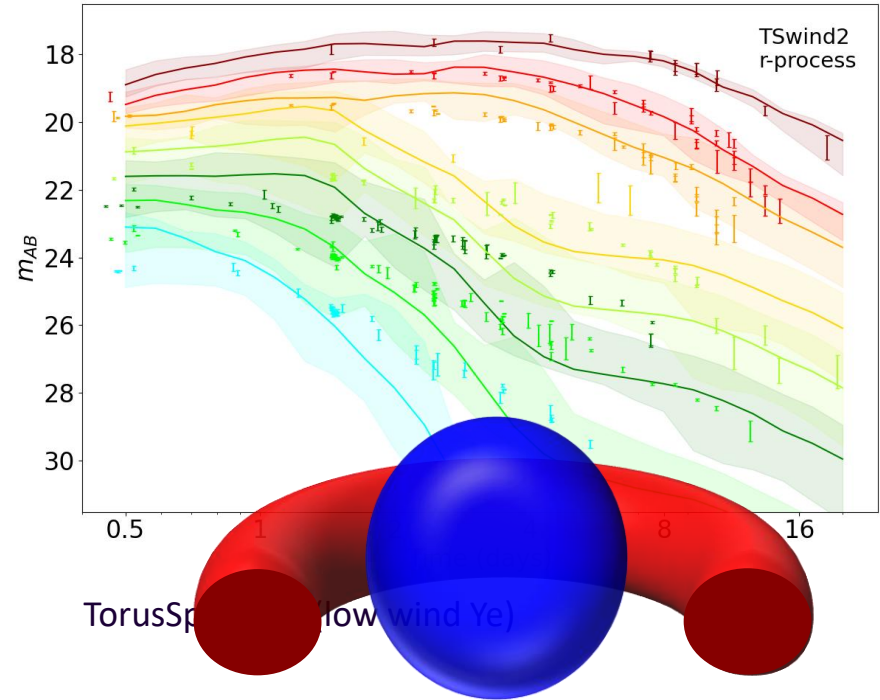
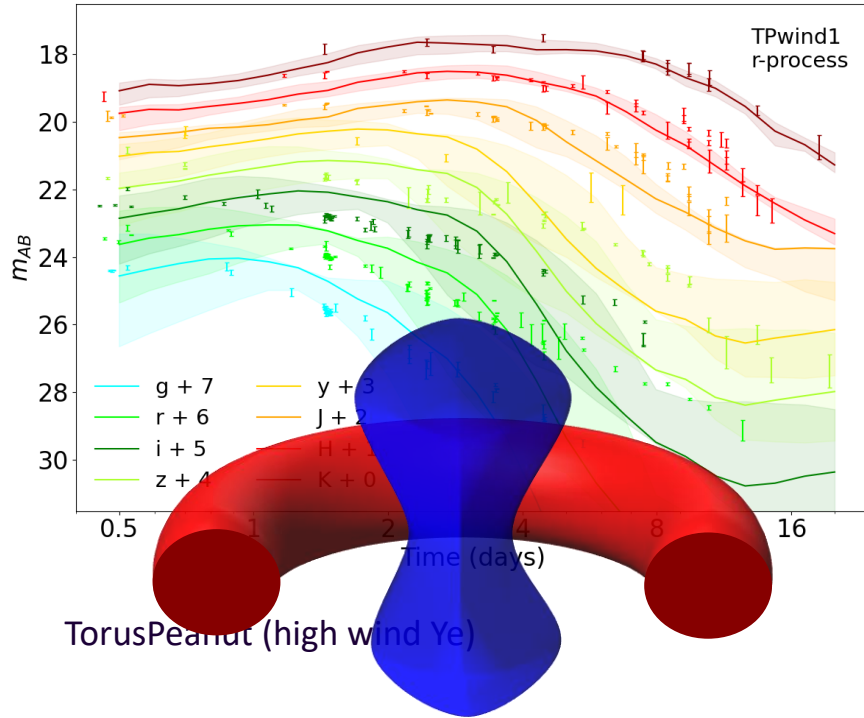


We construct a two component to describe the ejecta. One which describes low proton fraction matter, and one which has a high proton fraction matter.

# Radiative transfer model

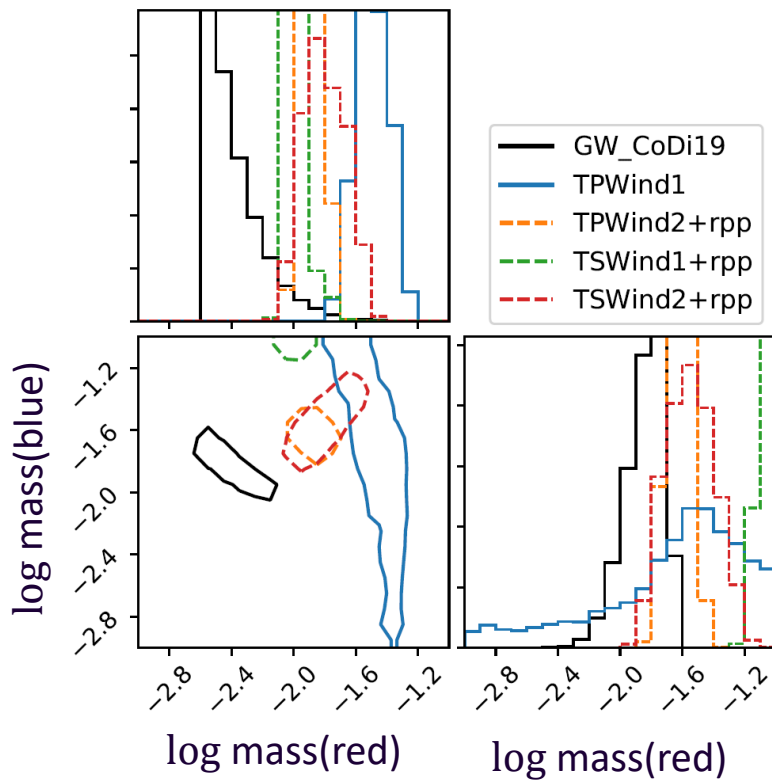
- Radiative transfer **SuperNu**. (Wollaeger et al 2013, 2014, Wollaeger, .., Kedia, Zenodo, 2023)
- Nucleosynthesis and heating from **WinNet**. (Winteler et al. 2012, Korobkin et al. 2012)
- Thermalization efficiencies (Barnes et al. (2016))
- Atomic opacities (Fontes et al. 2020)

# Kilonova Light curves

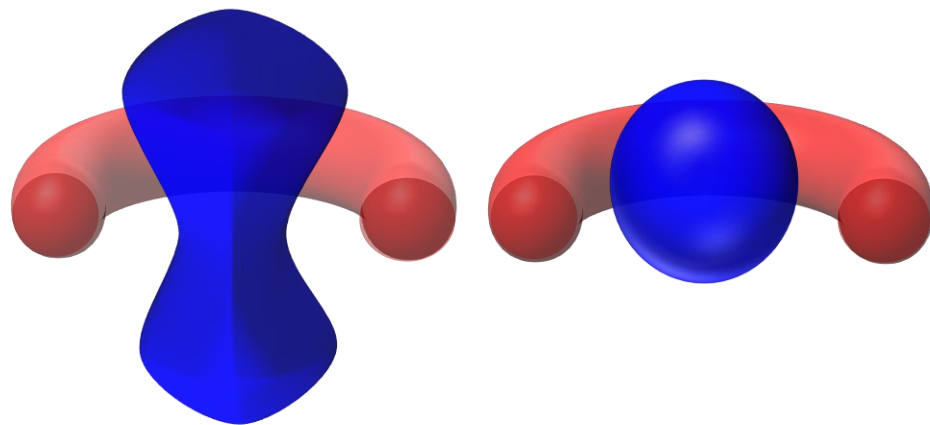


AK et al Phys. Rev. Research 5, 013168 (2023)

# Bayesian inference based ejecta masses from GW170817



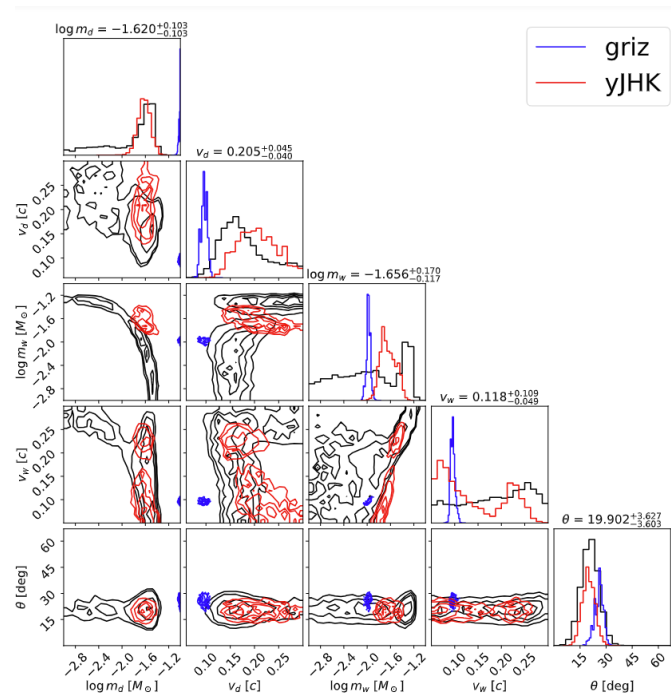
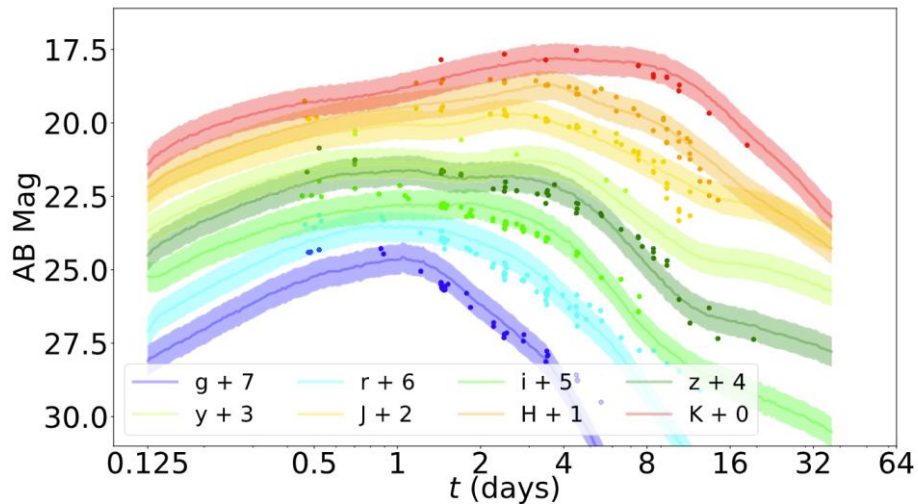
General agreement between the GW and kilonova observations on the inferred  $r$ -process enriched ejecta. However, how much mass was ejected depends on the morphology.



AK, et al, Phys. Rev. Res. 5, 013168 (2023);



# Light curves w/ Neural Networks



**Yinglei Peng\***, Ristic, Kedia, et al. [Phys. Rev. Res. 6, 033078 \(2024\)](#)

\*Conducted as undergrad at URochester, currently a graduate student at GeorgiaTech

Inference dependence on chosen wavebands.



# Outline

1. Introduction
  - Question: what is the astrophysical origin of *heavy* elements?
  - **Answer: Neutron star mergers -> Maybe**
2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)
3. Modelling Astrophysics: How much *r*-process is produced? (mass) (Answers: earlier slides)
4. Conclusions

# Future of observations and experiments

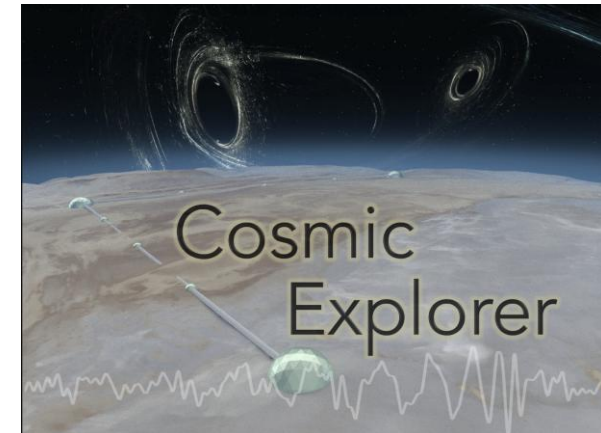
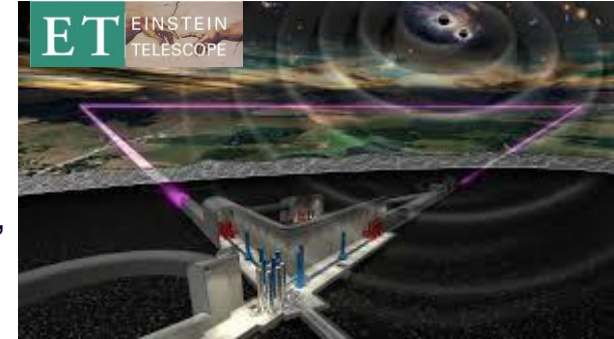


Nuclear experiments at:  
TUNL (Triangle), FRIB (MSU),  
ARIEL (TRIUMF)



EM observation:  
Nancy Grace Roman Space Telescope,  
Vera Rubin Observatory,  
James Webb Space Telescope

GW observation:  
LIGO-Virgo-KAGRA  
3<sup>rd</sup> Generation GW detectors



# Summary

- **Origin of elements**
- **Neutron star mergers — GW170817**
  - Supernovae
- ***r*-process nucleosynthesis**
  - Uncertainties in abundance from theoretical models
  - Isotopes that reduce uncertainties
- **Kilonova modelling**
  - Modelling in the two component (red, blue) method
  - Uncertainties in the amount of ejected mass produced.

# Outlook

- ***r*-process nucleosynthesis**
  - Constraints on nuclear reaction rates
  - Astrophysical environments: Supernovae, Neutron star mergers, Collapsars, etc.
- **Kilonova modelling**
  - Apply hydrodynamic data into kilonova model
  - Model asymmetries such as due to neutron star-black hole merger.
  - Nuclear uncertainties in radioactivity

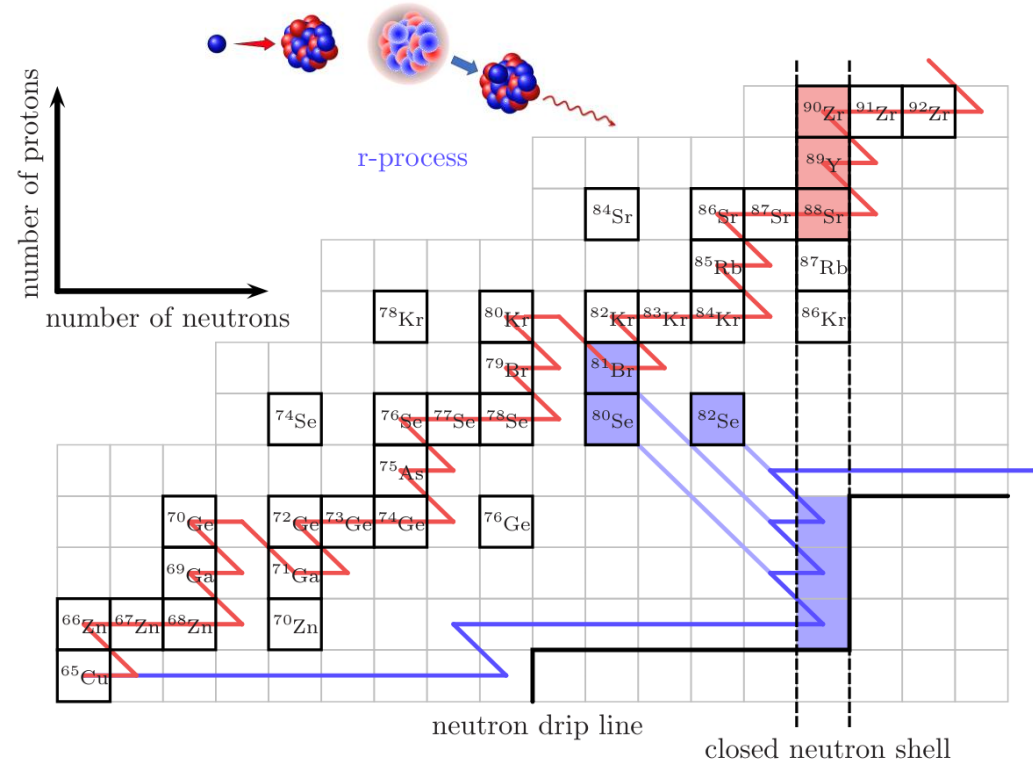
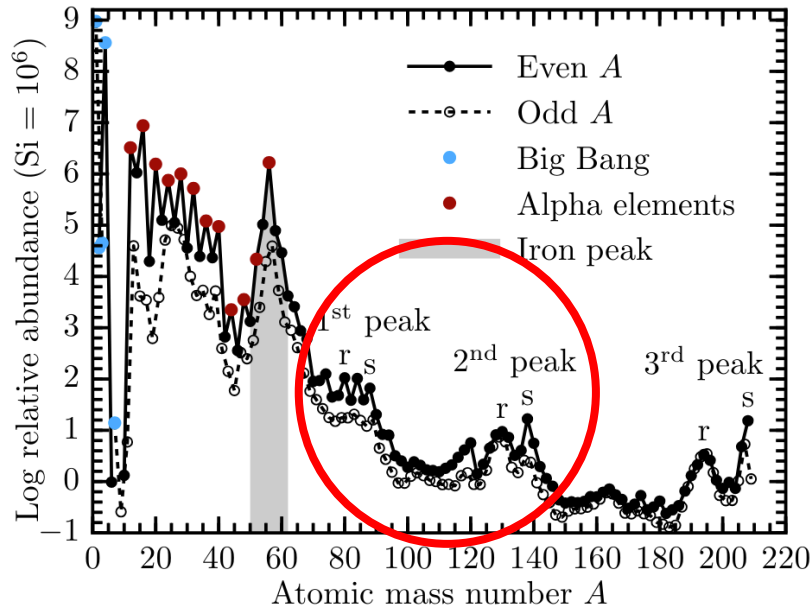
**More theory, observational, and experimental work in nuclear astrophysics.**



[askedia@ncsu.edu](mailto:askedia@ncsu.edu)

# Extra slides

# Solar chemical composition And the *r* process



Figures from [Lippuner \(2018\)](#)

*r*-process:  $\tau_{n\text{-capture}} \ll \tau_{\beta\text{-decay}}$

# Astrophysical sites for the $r$ process

## Binary Neutron star mergers:

(Meyer+1989, Metzger & Fernandez 14, Just+15, Radice+18, Kedia+22, Lund+24)

## Neutron star-black hole mergers:

Surman+08, Darbha+21, Curtis+23

## Magnetorotational supernovae:

Nishimura+15, Mosta+18, Reichert+21, Zha+24

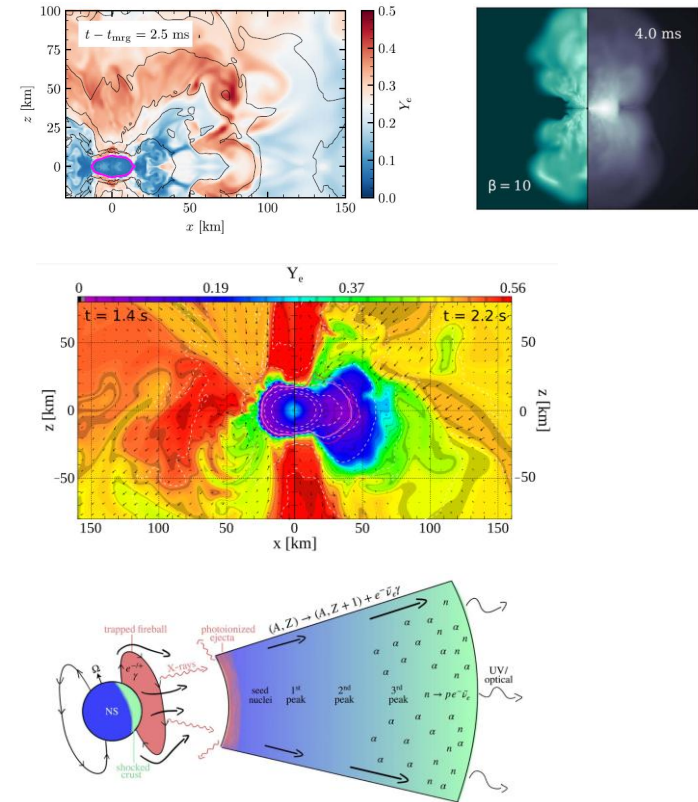
## Magnetar giant flares:

Patel+25a, 25b

## Collapsars:

Siegel+2019, Anand+2024

and **Others exotic scenarios**



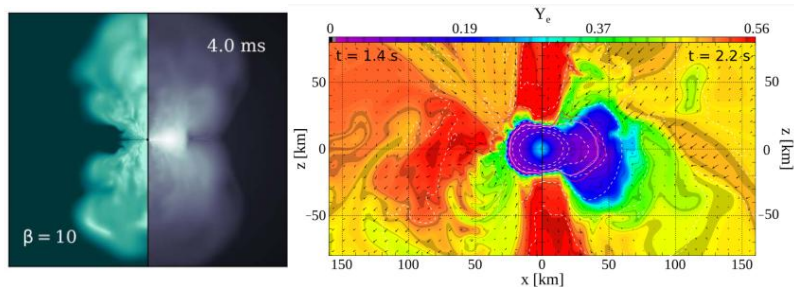


# Astrophysics and nuclear physics inputs

## Astrophysics:

Magnetized and non-magnetized disk winds from neutron star mergers  
(K. Lund et al 2024, Metzger & Fernandez 2014)

Magnetorotational supernova  
(Reichert et al 2021)

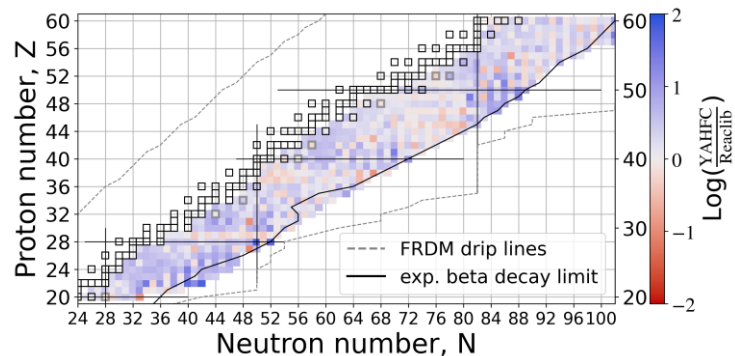


## Nuclear physics:

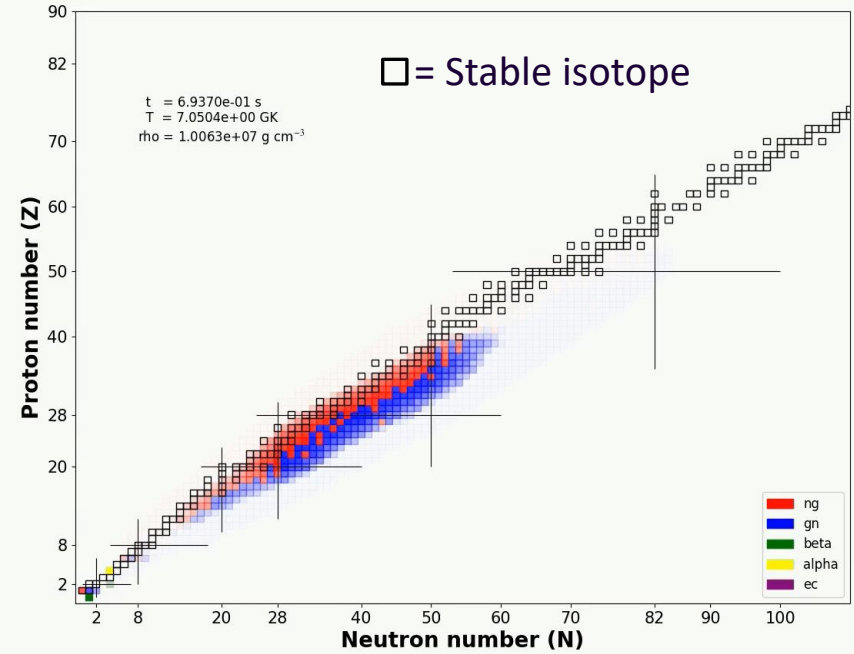
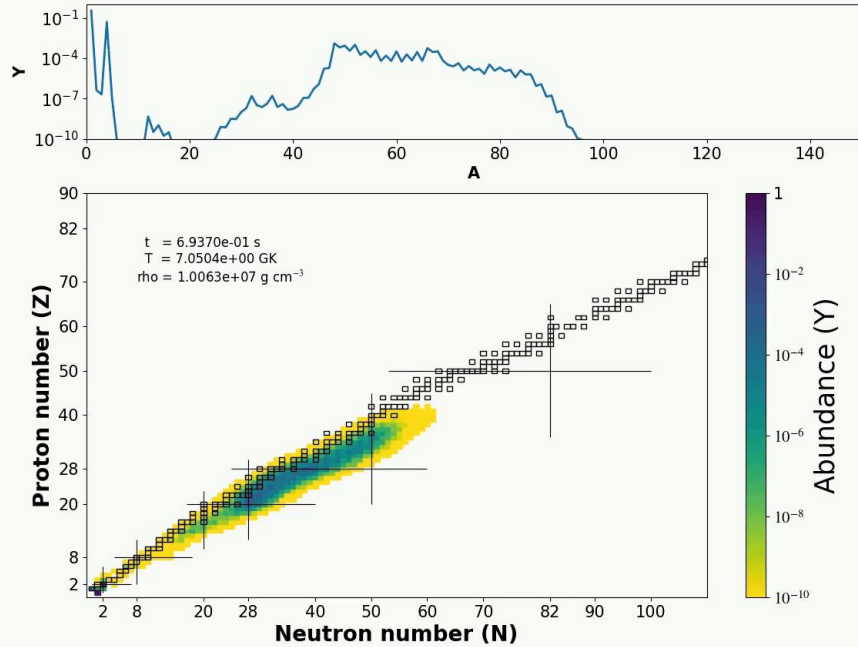
Neutron capture rates from YAHFC  
(Yet Another Hauser-Feshbach Code)  
(Ormand, Kravvaris, LLNL)

1000 isotopes,  $Z=20-60$ ;

Beta-decay rates



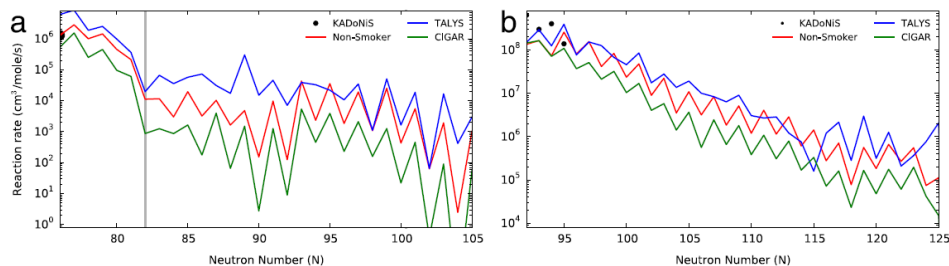
# R-process evolution



Movies by AK; Calculation done on **PRISM** (reaction network code)

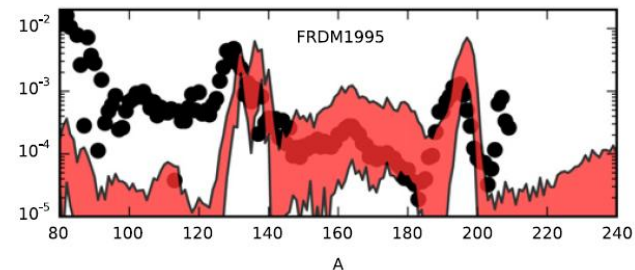
ng -> neutron capture  
gn -> photodissociation (neutron removal)  
beta -> beta decay

# Uncertainties in neutron capture reactions



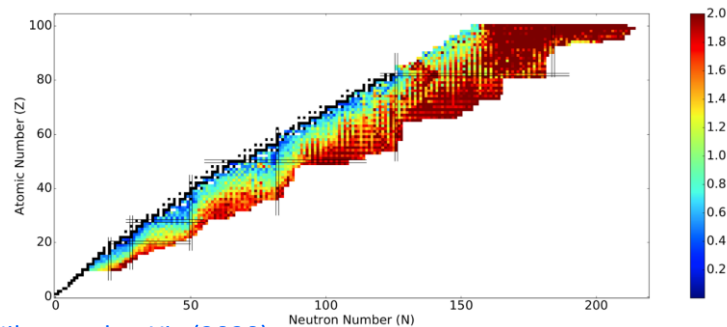
Sn –chain

Eu –chain



[Mumpower+, PNP \(2016\)](#) ;

multiplicative factors:  $10^{-3} - 10^3$  in n-cap rates;  
Mumpower et al ApJ 970:173 (2024)

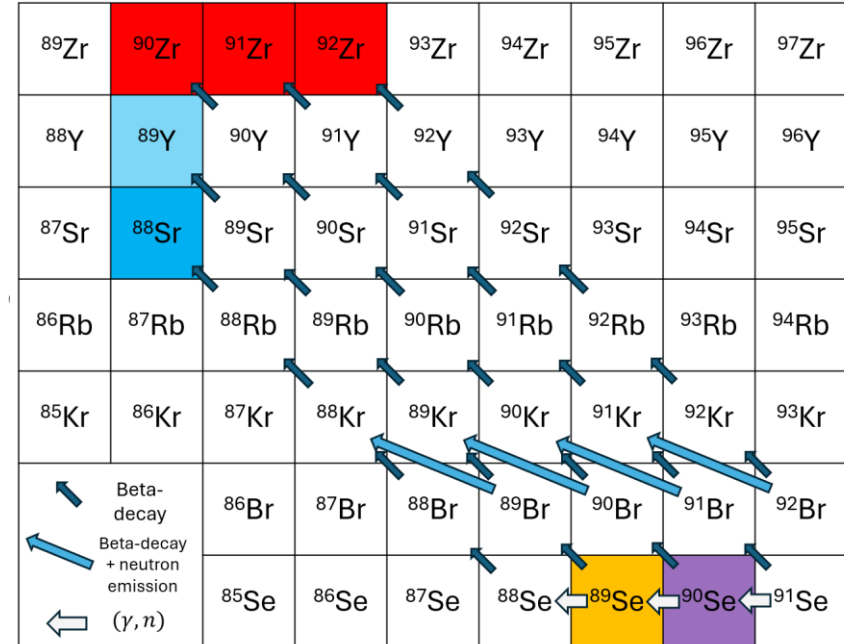
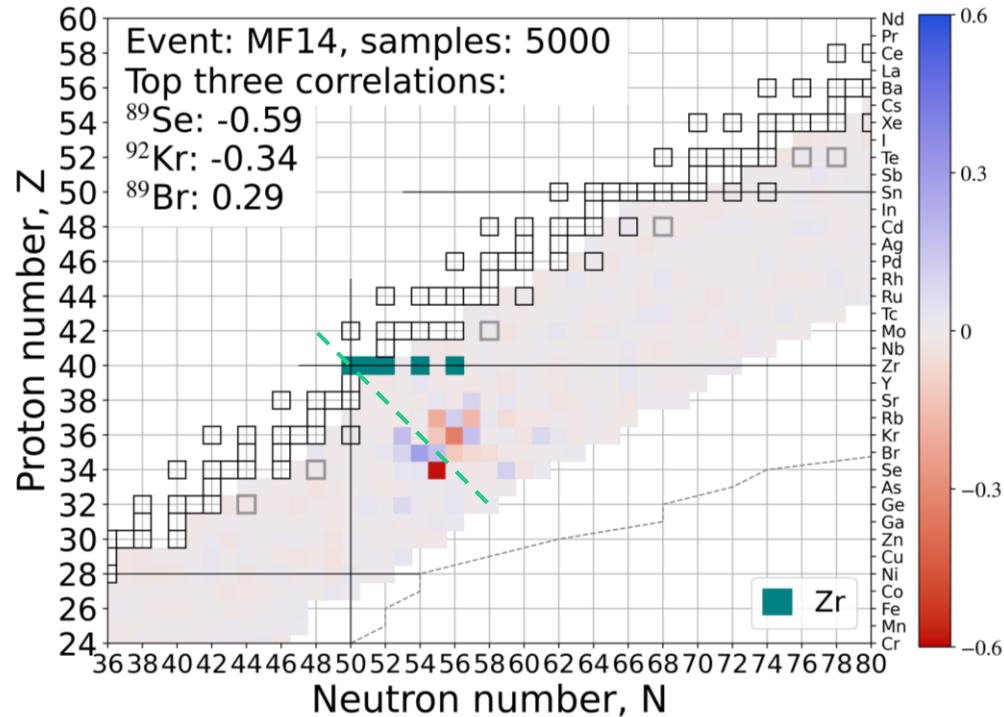


[Nikas et al, arXiv \(2020\)](#)

N-capture uncertainties

N-cap uncertainties propagated to abundances

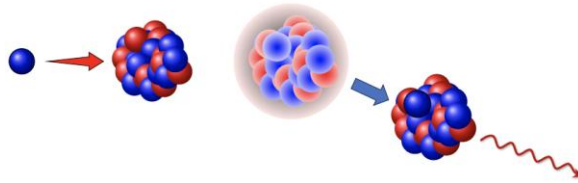
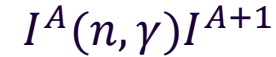
# Elemental abundance — isotopic neutron capture rate correlation



Due to detailed balance:  $\lambda_{\gamma, n}^{Z, A+1} \propto \lambda_{n, \gamma}^{Z, A}$

# Nomenclature: Reactions at play in $r$ -process

**Neutron Capture**



**Photodissociation**



**Beta-decay**



# Hauser-Feshbach statistical model

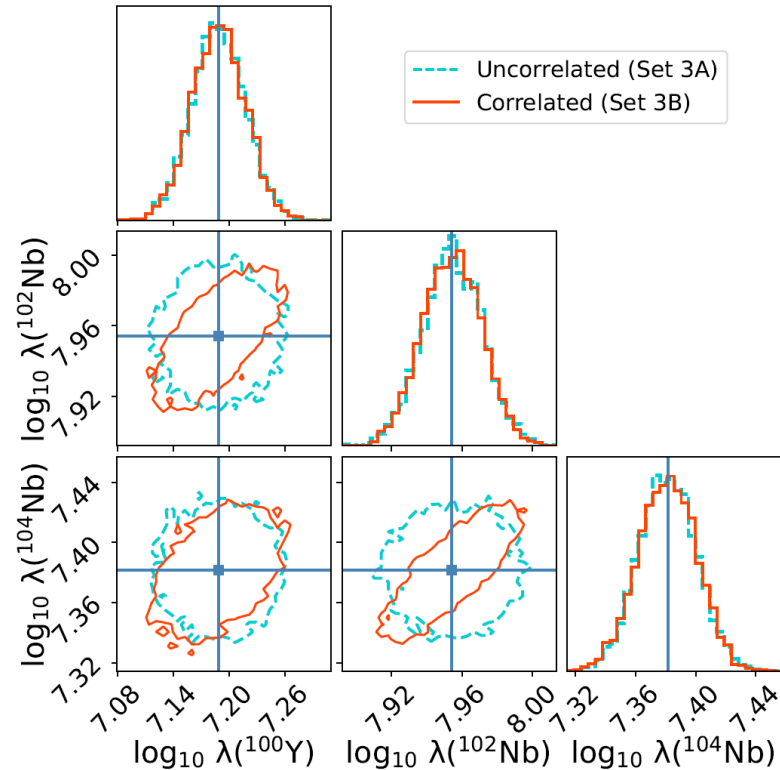
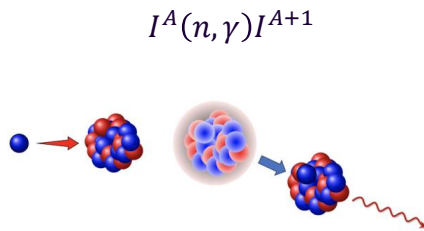
Hauser-Feshbach statistical model inputs:

☐ Mass Model

☒ Optical Model Potential

☐ Level Density

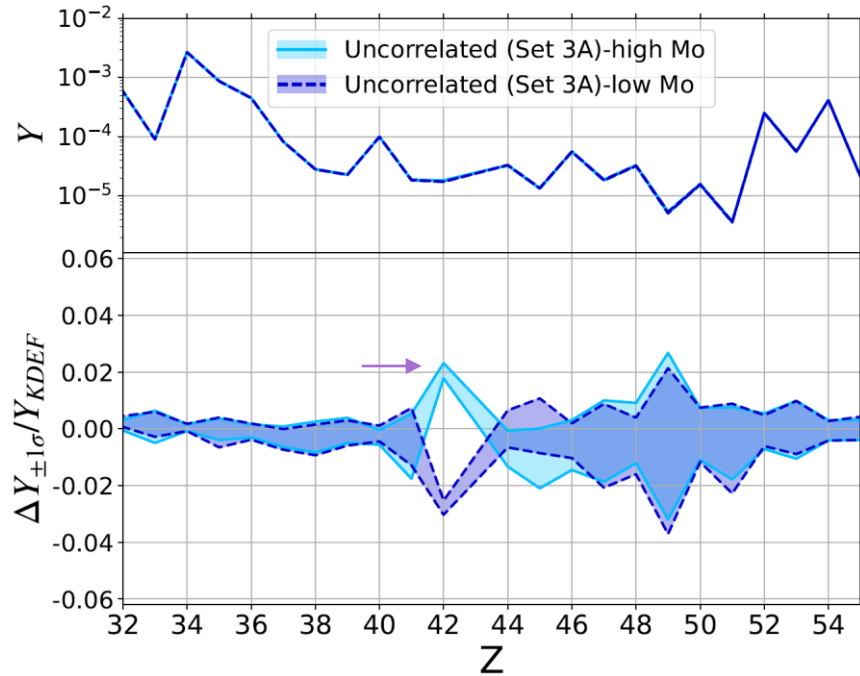
☐ Gamma-Strength Function



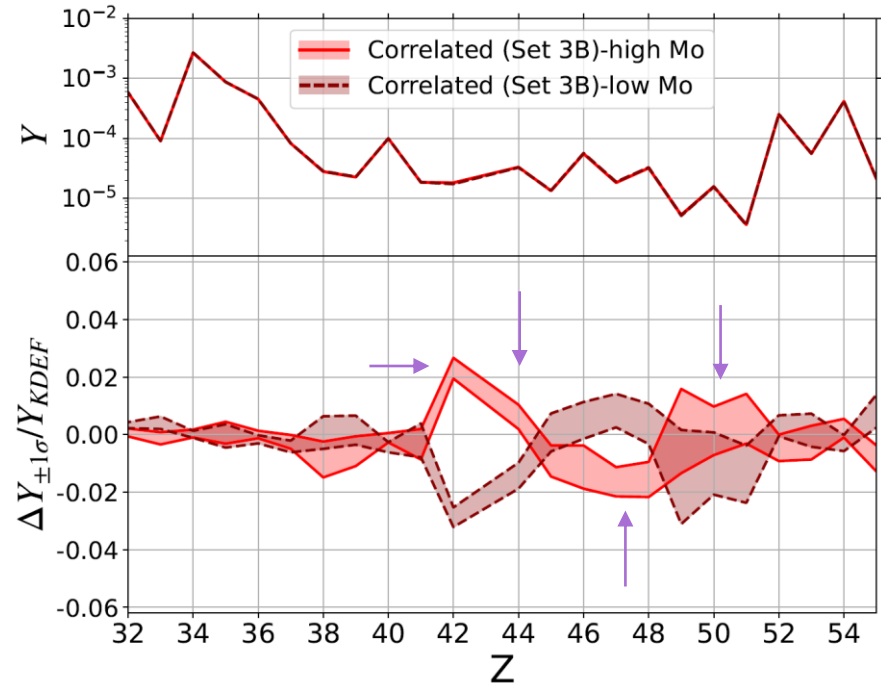
Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)



# Correlated neutron capture rates



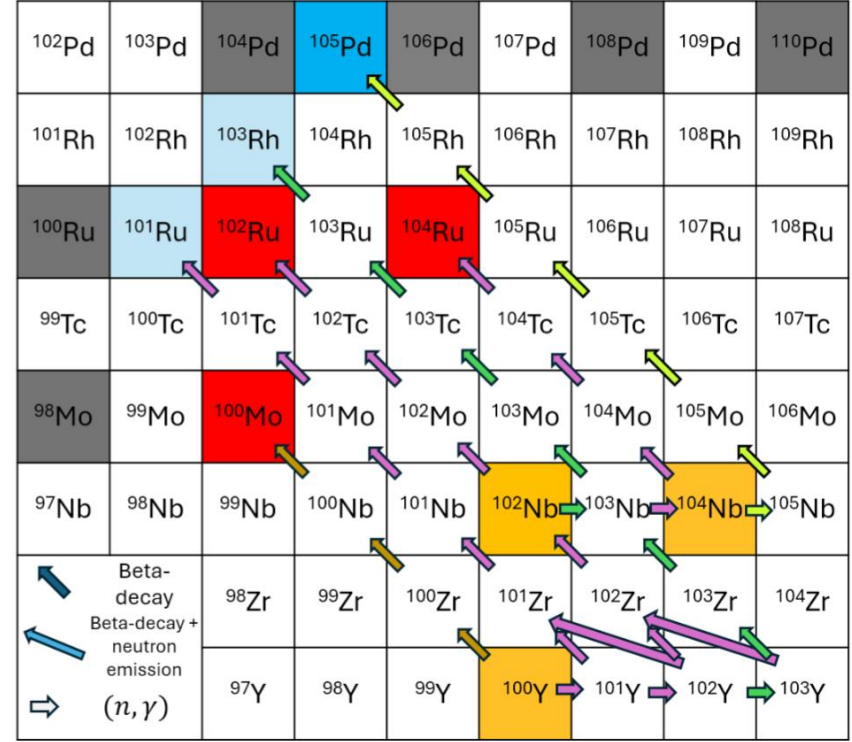
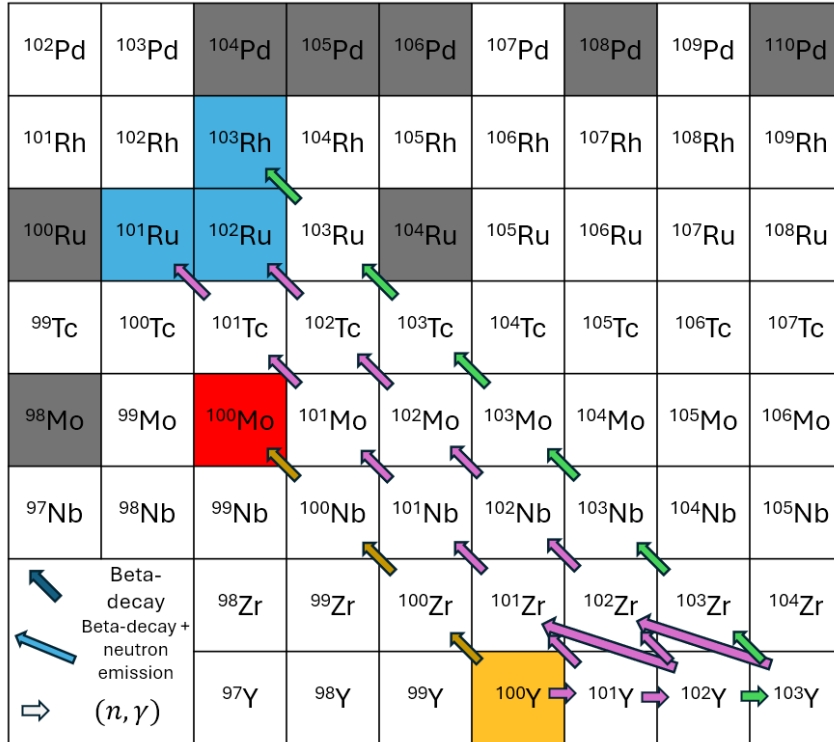
Uncorrelated: Simulations with high Mo (42) do not exhibit an increase or decrease in most other elements.



Correlated: Simulations with high Mo (42) also have high Ru (44), Sb(51) and low Rh (45), Pd (46), Ag (47), Cd (48).

Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

# Correlations in neutron capture rates – Toy model



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

# Supernova 1987a

First Multi-Messenger event

Distance  $\sim 50$  kpc

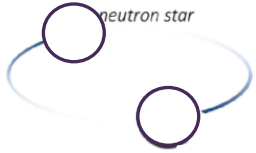
Electromagnetic (EM) and  
neutrino emission detected



# Bayesian inference of BNS ejecta

## Binary properties

$$\vec{x} = \{M_{\text{BIL}}, M_{\text{NS}}, \chi_{\text{BIL}}, \Lambda_{\text{NS}}, \dots\}$$

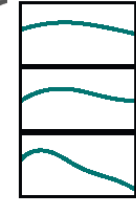
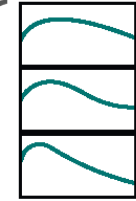


## Outflow properties

$$\vec{y} = \{M_{\text{dyn}}, M_{\text{wind}}, v_{\text{dyn}}, v_{\text{wind}}, \dots\}$$



## Light curves



Fit formulae to  
numerical simulations

$$\vec{x}_1$$

BNS merger modelling  
(Numerical relativity)

Radiative transfer  
simulations

$$\vec{y}_{11}$$

$$\vec{y}_{12}$$

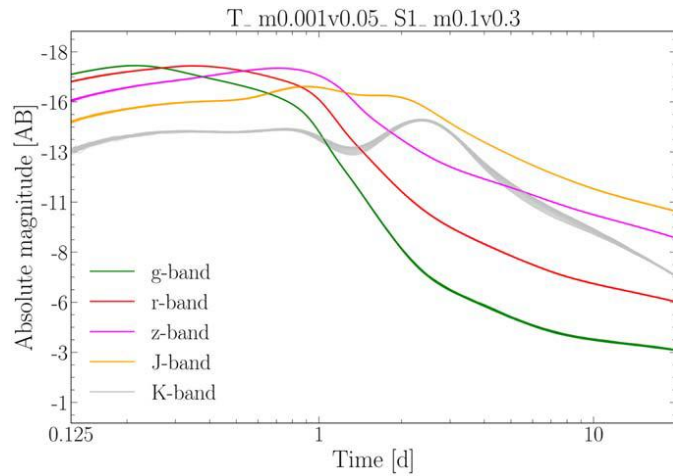
$$\vec{y}_{13}$$

$\vdots$

Ejecta material that  
undergoes  
nucleosynthesis

Kilonova  
modelling

Adapted from Raaijmakers et al, [ApJ \(2021\)](#)



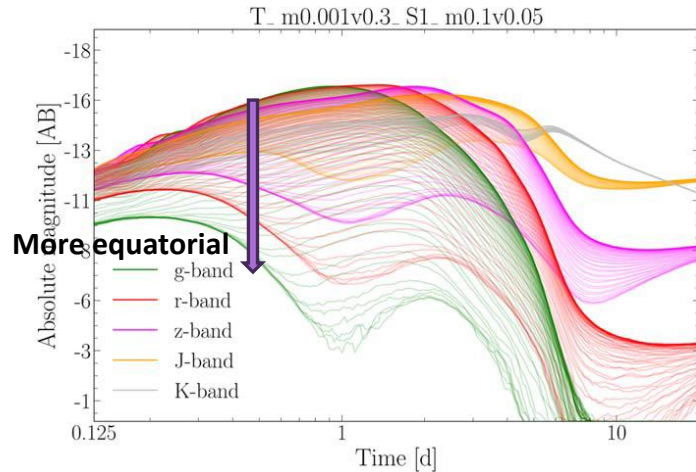
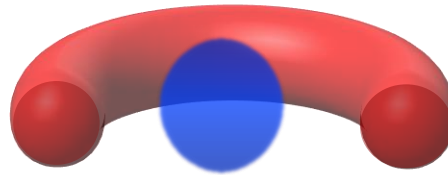
# Lanthanide curtaining

## A smoking gun for r-process

Curtaining of low wavelength (“blue”) light by Lanthanides (present in the “red” ejecta) produced by the nucleosynthesis.

Top panel: Ejecta more spherically symmetric and less obstructed by the “red” component. Therefore, there is virtually no angular dependence.

Bottom: “Red” component travels further and obstructs light emitted along the equator. Therefore, Lanthanide curtaining is observed. Low wavelength light is suppressed substantially along the equator. High wavelength light passes unaffected (spherically symmetric).

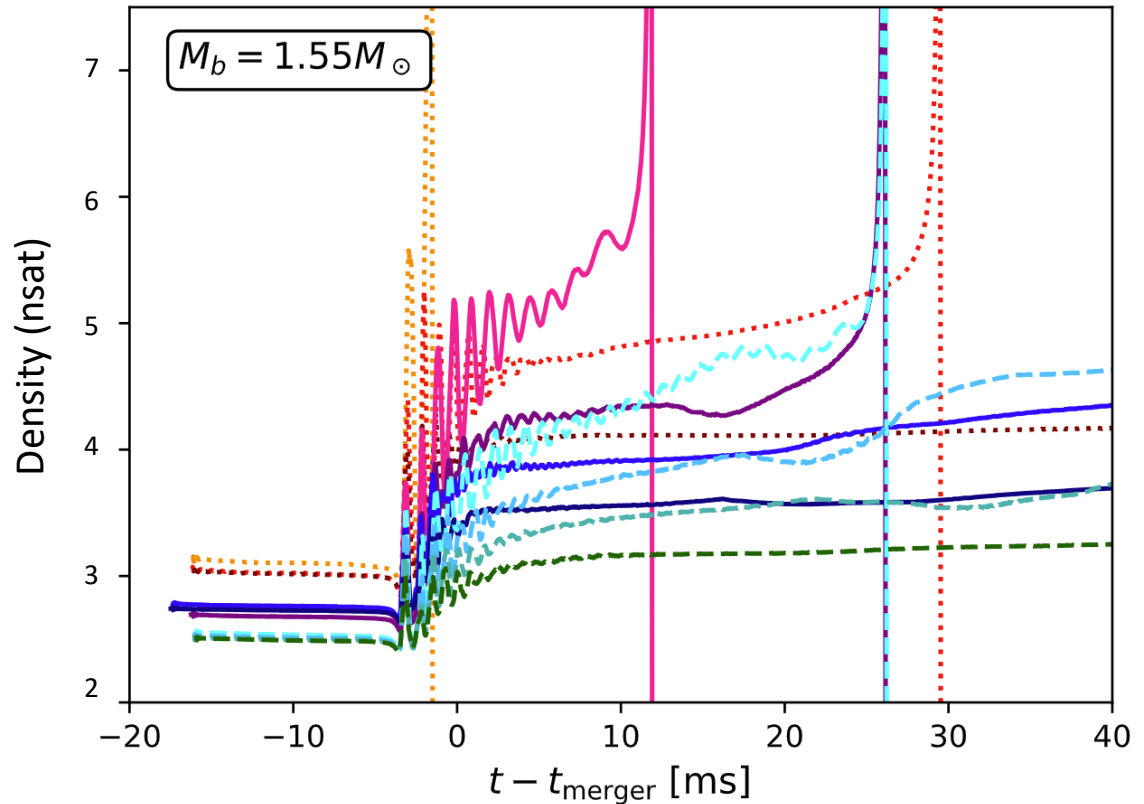


Wollaeger et al ApJ 2021; See also Korobkin+2021, Kasen+2015

# Collapse of hypermassive NSs

Postmerger hypermassive  
Neutron star remnant  
dependence on the  
nuclear Equation of State  
(i.e. what defines the  
neutron star structure)

Vertical drop represents  
black hole formation.



Wei Sun, Mathews, Kedia, et al., (in prep)